Applied Optoelectronics (AAOI)

Full investment research | Model assumptions and source evidence

Full investment research and current operating model | October 19, 2026 | Framework 8.0

Current investment view

AAOI can earn substantially more as qualified high-speed module shipments grow, its product allocation shifts toward 1.6T, and external light-source modules become commercial sales. The forecast depends on customer acceptance, available DSPs and lasers, repeatable manufacturing, and unit-cost improvement. Installed factory capacity earns returns only after those steps succeed.

The current **Q4 2027 modeled ceiling is about $463**, a decrease of **less than 1%**. The nearer **Q4 2026 ceiling falls about 4%, from $247 to $236**. The difference follows the earnings windows: the new risk primarily affects earlier quarters, and the active Q4 2027 window includes three unchanged 2028 quarters. A rounded unchanged headline price does not erase the exact reduction stored in the model.

Analytical Base: 70/100, from 42 points out of 60. Protected Excel Base: 72/100. The Cycle score falls from 10 to 9 because new owner-supplied analyst notes raise the risk of slower qualification-to-volume conversion. The other categories and five existing risk deductions remain. Valuation stays at **20x forward-four-quarter modeled operating profit**, using the existing starting-quarter economic-share schedule and USD convention.

Numbers in prose, charts and tables are rounded to whole units. Technical product names such as 1.6T retain their standard notation. Complete calculation precision remains in the synchronized model JSON and workbook.

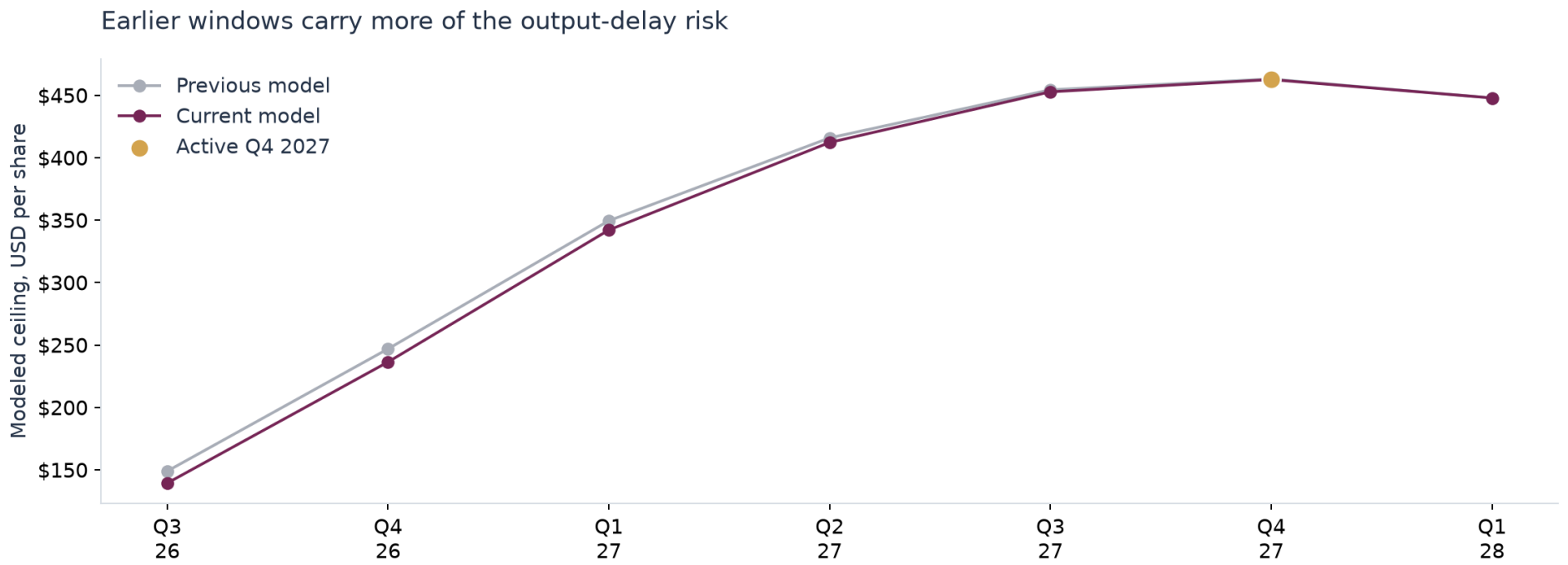


Figure 1. Current and previous quarterly ceilings. The gold point marks the active Q4 2027 horizon. Company valuation method and share schedule are unchanged.

Ceiling quarter	Previous USD/share	Current USD/share	Revision
Q3 2026	149	139	-6%
Q4 2026	247	236	-4%
Q1 2027	350	342	-2%
Q2 2027	416	412	Down less than 1%
Q3 2027	454	453	Down less than 1%
Q4 2027	463	463	Down less than 1%
Q1 2028	448	448	0%

Analytical Base and conviction

Category	Score	Evidence and limitation
Core Thesis	14/20	Internal lasers and US manufacturing support execution; standardized modules remain competitive.
Cycle Stage	9/10	800G output is ramping, but current analyst production notes justify a bounded delay state.
TAM Expansion	8/10	Higher bandwidth and external light sources broaden addressable demand; avoid adding overlapping components.
Catalysts	5/5	Qualified Texas output, high-speed mix, accepted ELSFP and dated customer conversions.
Macro	9/10	AI networks require more bandwidth and optical links, subject to deployment timing.
Alpha	2/5	The edge depends on qualified output, product costs and timing against a demanding existing forecast.
Risks	-5	Execution/components; competition/concentration; ELSFP qualification; funding/dilution; architecture/legacy demand.
Total	42/60 = 70/100	Analytical score. Excel Base remains 72.

The one-point Cycle reduction does not create another risk deduction for the same production issue. Base describes thesis quality. It does not multiply operating profit or the ceiling. A stronger score would require observed output and customer acceptance against the dated ramp; weaker qualification, financing or unit economics can justify another targeted review.

Current 800G qualification case and source decisions

The owner supplied four translated analyst-note images and confirmed that the note was written October 18, 2026. The analyst's identity is unknown. Excerpt points 6-13 are available; points 1-5 and the original report are missing. The source reports high equipment utilization alongside much lower good 800G output. The exact factory, customer, SKU, observation date and denominator are still unresolved. This is credible enough to price a bounded execution risk, while its scope limits the selected outcome.

Selected probability: 25%. Conditional outcome: 800G saleable conversion runs one quarter behind its existing reference during Q4 2026 through Q4 2027. Only conversion changes. Allocated capacity, ASP, 1.6T, ELSFP, other products, shares and other accepted branches are retained. The 2028 reference resumes, with no invented catch-up sales and no second permanent cancellation haircut. The source is a single unnamed analyst and the official earlier ramp supports the reference, so the case remains a limited adverse state rather than a wholesale replacement forecast.

The reference already includes the inherited 30% chance of five extra percentage points of utilization from Q1 2027, capped at 95%. The new 25% probability applies to the incremental lag relative to that saved reference. It does not add another shortage to both generations. One common adverse state spans the affected quarters; these are correlated observations, not separate quarterly confirmations.

Expected 800G conversion equals 75% of the current reference plus 25% of the previous quarter's reference conversion. Lost sales equal allocated monthly capacity times three months, the conversion reduction and retained ASP. Of the COGS attached to delayed units, 85% is avoided and 15% remains as a labeled fixed-cost/underabsorption proxy. Opex is retained because engineering, ramp support and staff remain. OP loss equals lost revenue less avoided COGS. No further idle-cost or rework deduction is stacked. The retained-cost fraction is an analyst estimate, with 0%, 15% and 25% sensitivities shown below.

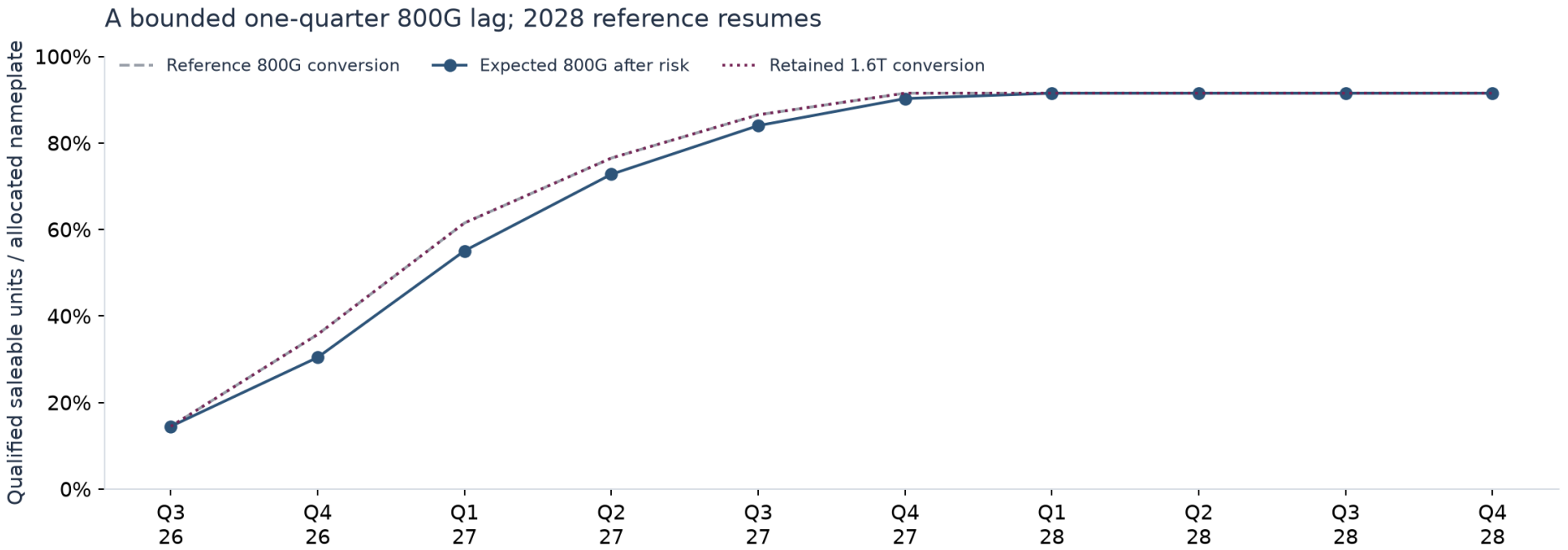


Figure 2. Qualified saleable output conversion. The current case affects 800G only; the dotted 1.6T reference is retained. Equipment utilization and optical-chip yield have different denominators.

Quarter	Reference 800G conversion	Conditional lag conversion	Expected 800G conversion	1.6T conversion
Q3 2026	14%	14%	14%	14%
Q4 2026	36%	14%	30%	36%
Q1 2027	62%	36%	55%	62%
Q2 2027	76%	62%	73%	76%
Q3 2027	86%	76%	84%	86%
Q4 2027	92%	86%	90%	92%
Q1 2028	92%	92%	92%	92%

Quarter	Reference 800G conversion	Conditional lag conversion	Expected 800G conversion	1.6T conversion
Q2 2028	92%	92%	92%	92%
Q3 2028	92%	92%	92%	92%
Q4 2028	92%	92%	92%	92%

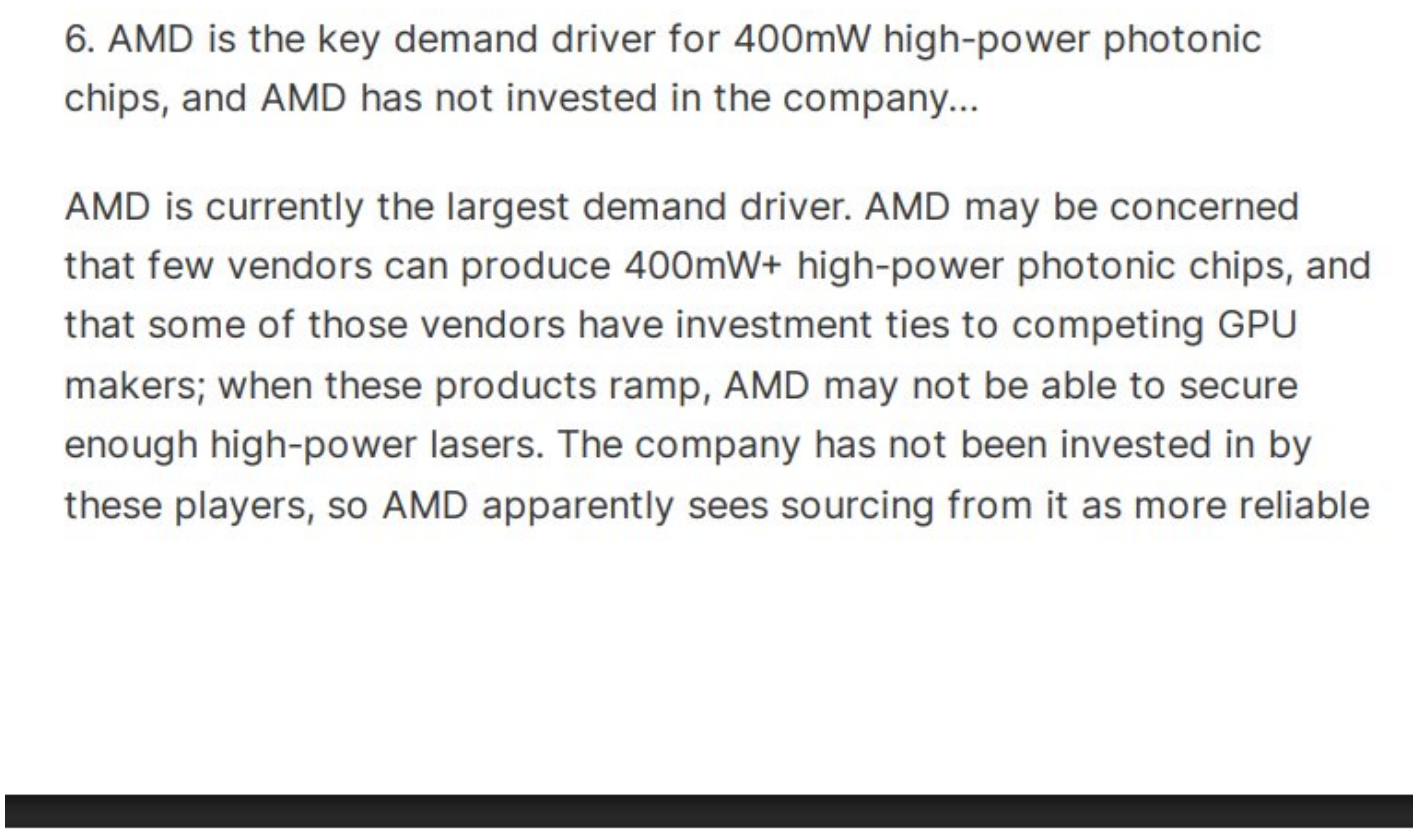
Quarter	800G revenue reduction, USD M	Avoided COGS, USD M	Retained COGS, USD M	OP reduction, USD M
Q4 2026	38	21	4	16
Q1 2027	41	23	4	18
Q2 2027	22	12	2	10
Q3 2027	13	7	1	6
Q4 2027	5	3	0	3

Individual excerpt claims and overlap

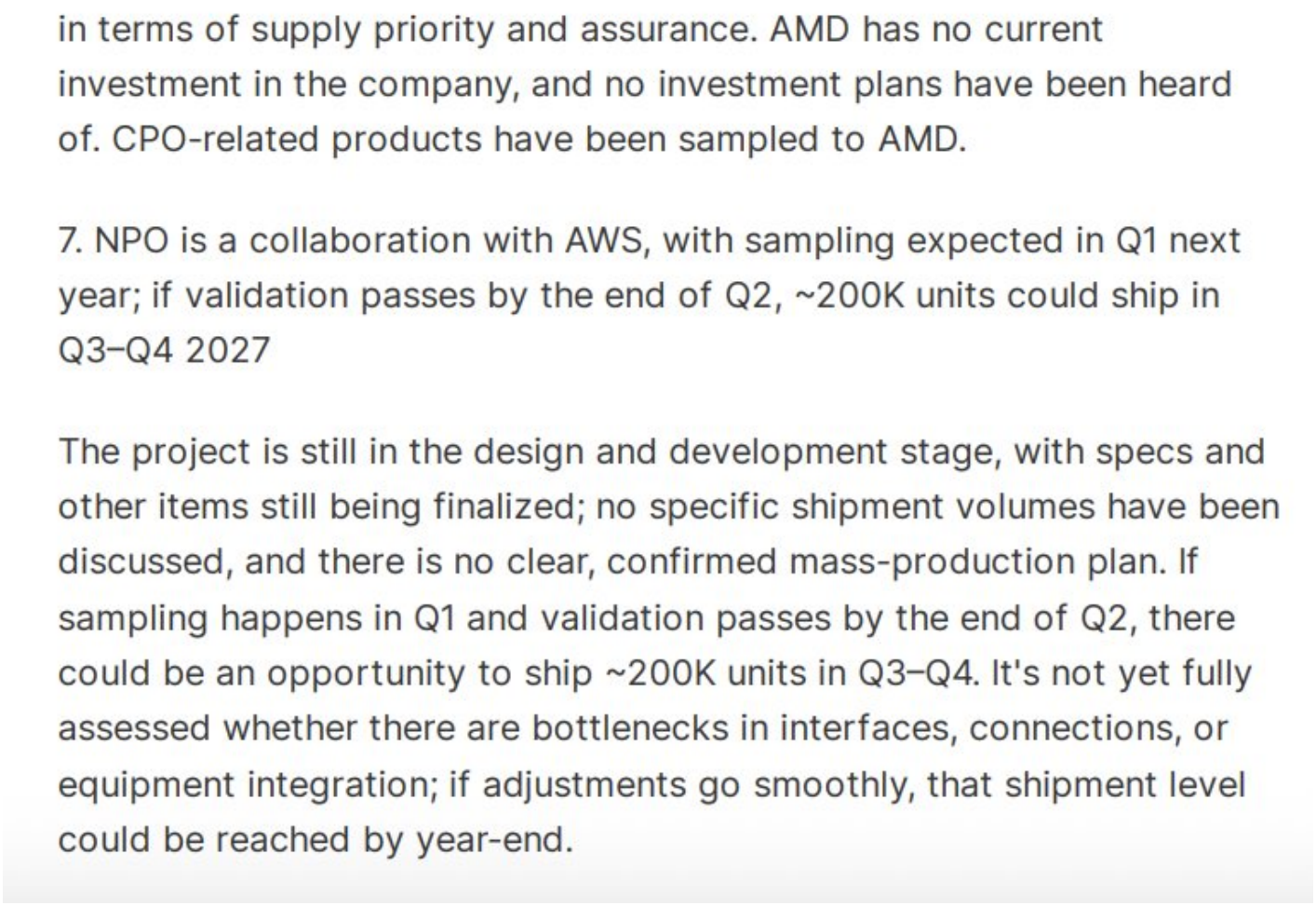
Excerpt	Decision	Reason and missing evidence
P6: AMD high-power samples	No additional revenue	Samples have no accepted customer volume, price or delivery date; internal chips and external ELSFP sales must remain separate. No AMD investment is modeled.
P7: AWS NPO schedule and possible 200,000 units	No extra units and no all-company cap	2027 interpretation follows the owner-confirmed note date. An AWS NPO unit is not mapped to an ELSFP module; customer share, modules per system, costs and acceptance are missing. Existing all-customer ELSFP output already covers a broader pool.
P8: DSP allocation and supply	Relevant execution context	MRVL/AWS and Oracle routes do not identify additional lost accepted units beyond the selected 800G state. New customer-specific delays could change the bounded case.
P9: InP supply risk	No separate blanket haircut	Substrate, epitaxy, die qualification and module acceptance are different stages; no matched incremental AAOI supply loss is quantified.
P10: 65% policy interpretation	Existing policy branch retained	Non-mainland wording differs from the old US-value interpretation. Original legal scope is unresolved; a repeated policy chain adds no second branch.
P11: Amazon purchase/warrant description	Existing contract inventory retained	Original full-vesting denominator is $4 billion; an excerpt about $400 million cannot establish a new contract or repeat the warrant dilution.
P12: revenue, customer mix and ASP	No imported numerical inputs	The excerpt contains inconsistent currency/scale labels and unstated customer-share periods. An original dated table is required.
P13: low good 800G output	25% bounded one-quarter lag	Captures the incremental execution residual above the saved low near-term conversion. No double multiplication of a good-output/nameplate ratio by mechanical utilization.

These zero additional increments follow specific overlap or unit/contract gaps after comparison with the saved model. They do not assign zero likelihood merely because a statement lacks official confirmation. A full company review and rescore is completed under the owner's current instruction, even where a target changes only slightly.

Supplied analyst originals



Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Points 6-8: high-power samples, NPO timing and DSP sourcing. Original exhibit, panel 1 of 2; read from top to bottom.

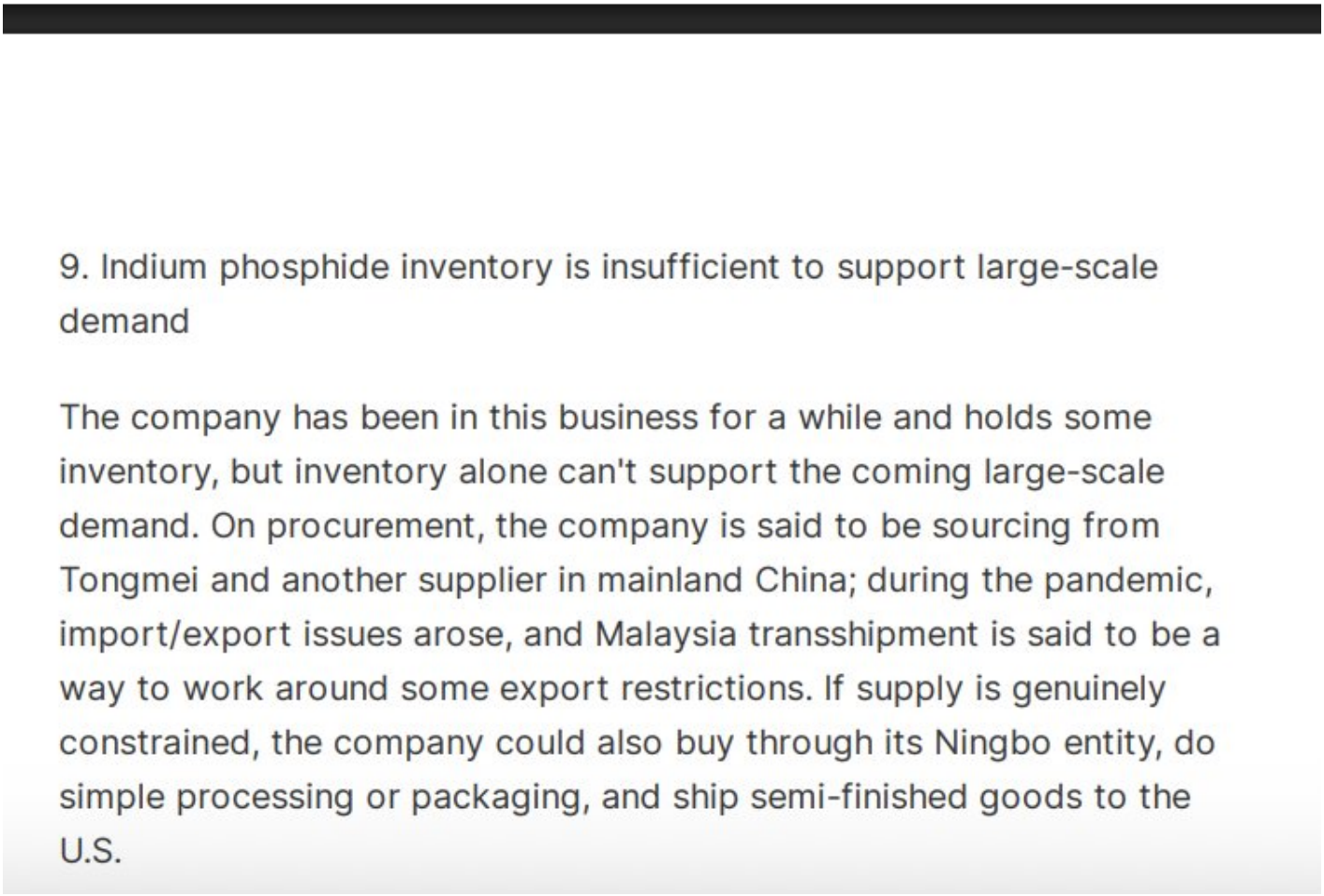


Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Points 6-8: high-power samples, NPO timing and DSP sourcing. Original exhibit, panel 2 of 2; read from top to bottom.

8. AWS's 800G DSPs are allocated by Marvell through the end customer on an order basis and typically aren't in short supply; 1.6T Broadcom and Marvell DSP supply and lead times remain tight

DSPs for AWS supply aren't in short supply, since those products use Marvell chips and AWS has an arrangement with Marvell under which AWS allocates DSPs. Oracle-related demand requires the company to negotiate directly with suppliers; some advance orders have been placed, and 800G looks fine for now. For 1.6T there are two DSP options — Broadcom and Marvell. Broadcom supply isn't great either and is broadly tight, but the company's current orders appear manageable and some supply has already been secured. Marvell's 1.6T DSP lead times are also unclear; only Cambridge Technology, which prepaid to lock in orders last year, is said to have relatively clear supply, and Marvell's newly launched products may be prioritized for Cambridge. Credo and Macom second-tier DSPs are a backup path for white-box products.

Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Points 9-11: supply, policy and customer/warrant commentary. Original exhibit, panel 1 of 2; read from top to bottom.

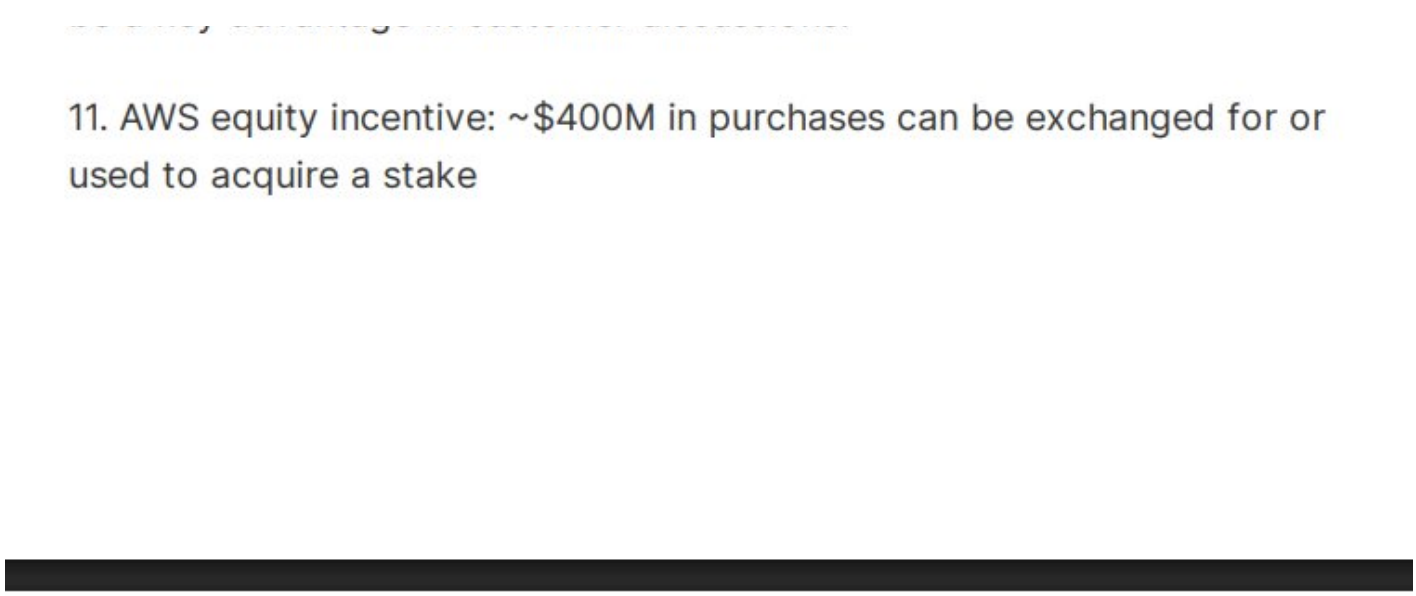


Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Points 9-11: supply, policy and customer/warrant commentary. Original exhibit, panel 2 of 2; read from top to bottom.

10. The FCC's 65% materials rule is more likely to focus on the non-mainland-content share; U.S. domestic manufacturing becomes a competitive advantage in winning orders

If the FCC imposes such requirements, CSPs will likely progressively require module makers to use as much overseas content as possible; for chips with both mainland and overseas options, they'll likely be asked to choose overseas. If overseas content in the finished module is high enough, manufacturing in mainland China should still be possible; more conservatively, manufacturing in Southeast Asia should be feasible. Very few vendors currently manufacture modules in the U.S., and one alone can't meet supply, so overseas production should be permitted — the constraint is more likely to be the non-mainland content share. Once 800G and other products ramp, U.S. factory output could exceed mainland China and Taiwan combined; U.S. domestic manufacturing capability means greater supply-chain security and can be a key advantage in customer discussions.

Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Point 12 and related customer, revenue or product statements; original units require clarification. Original exhibit, panel 1 of 2; read from top to bottom.



There is such an arrangement, apparently with AWS. Once purchases reach roughly $400M, a stake can be exchanged for or acquired, though the exact size of the stake is unclear.

Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Point 12 and related customer, revenue or product statements; original units require clarification. Original exhibit, panel 2 of 2; read from top to bottom.

12. Customer mix: Microsoft ~20%, AWS ~20%, Oracle unspecified; AWS remains the main growth driver next year

Microsoft accounted for ~20% last year and is estimated at ~20% this year. Total company sales this year are expected at just over RMB 1B to RMB 1.x billion. Microsoft's 100G volume is estimated at ~1M units at an average price of 100M; Microsoft also has some 400G and AOC products. Within total company revenue, CATV is likely just over RMB 300M, or ~35%; of the remaining ~65%, Microsoft is likely 15–20%, and if Microsoft is 20%, Amazon is likely around 20% as well. Oracle's exact share hasn't been estimated. AWS may want first-tier suppliers to shift more toward high-end products, gradually reduce their 400G share, and leave some 400G share to the company; AWS wants the company to become one of the primary supply groups for one "2000" spec 800G product.

13. The 800G line is still ramping; actual output released may be only 10–20% of equipment nameplate capacity

800G utilization may be around 90%, but the line is still ramping, and there's still a gap between stable production capacity and equipment nameplate capacity — actual output released may be only 10–20% of equipment capacity. Strictly speaking, it's not yet a mature capacity utilization situation; the company is still in expansion and ramp-up. Low-end 100G is likely still running at close to 100% utilization; 400G is also near full capacity and should stay close to 100%, with some expansion needed, though not a large one.

Original owner-supplied analyst excerpt, note date October 18, 2026 per owner. Analyst identity unknown. Point 13: equipment utilization and good 800G output; the source denominator is unresolved.

InP stage comparison and competing evidence

P Equity Research's October 19 expert-call article contains a China-based speaker whose name and background are undisclosed; the call is dated only October 2026. It estimates about 10,000 raw conventional EML die sites per three-inch substrate and a little over 2,000 usable chips after downstream losses. A separate six-inch example says fewer than ten of 100 grown crystals qualify for six-inch finished wafers, while most remaining crystals are downgraded to smaller wafers. Neither measurement establishes AAOI's finished 800G module yield.

The call's 200-300 mW CW die-area and stricter substrate-defect discussion supports evaluating qualified high-power supply. It does not establish acceptance or failure of AAOI's 400 mW program. The assumed CPO count of eighteen CW sources is architecture-specific, with no universal NPO-to-ELSFP mapping. The expert expects supply to ease by the second half of 2027; the publisher explicitly disagrees and expects longer tightness. Capacity and yield are contested, with product grade, size equivalents, customer qualification, red-phosphorus inputs and downstream MOCVD availability as separate constraints. No second permanent AAOI cost or volume shock is added from these broad statements.

Q2. How is the InP substrate consumption of a single optical module calculated?

Optical modules are mainly divided into two technical routes, EML and CW. Taking a 3-inch substrate as an example, without considering downstream epitaxy and optical chip yield, one substrate can be cut into about 10,000 EML chips for 100G and 200G. But once downstream yield is included, the effective chip output per wafer is a little over 2,000. For conventional 70 mW and 100 mW CW chips, one 3-inch substrate can be cut into about 8,000 dies, and after yield the effective output is likewise a little over 2,000. Although CW chips have fewer dies per wafer, their overall yield is usually slightly higher than EML's.

Q3. How will CPO solutions affect demand for InP substrates?

CPO is currently at the sample stage and not yet in large-scale volume production. CPO will drive significant growth in InP substrate demand, mainly through die area and quality requirements. **A CPO module is**

Original InP expert-call PDF page 7. Raw laser-die sites, effective good chips and architecture-specific CW counts are distinct from AAOI module-line conversion. Source: P Equity Research, October 19, 2026; anonymous expert, October call. Original exhibit, panel 1 of 2; read from top to bottom.

expected to use 18 CW light sources, with power requirements far above conventional products. For example, the 200 mW and 300 mW CW chips being developed by Yuanjie Semiconductor have a single bare die area 2 to 3 times that of today's 70 mW and 100 mW chips, which directly increases substrate area demand. CPO modules also place stricter demands on substrate quality. Because continuous emission from CW lasers generates serious heat, high defect density raises the probability of chip failure. Therefore, for 300 mW CW sources used in CPO, **the maximum substrate defect density is required not to exceed 200**, whereas for orders for conventional 100G and 200G optical modules or 70 mW and 100 mW CW sources, **the average defect density requirement is only below 500**.

Q4. How will market supply meet the demand growth expected in 2027?

Original InP expert-call PDF page 7. Raw laser-die sites, effective good chips and architecture-specific CW counts are distinct from AAOI module-line conversion. Source: P Equity Research, October 19, 2026; anonymous expert, October call. Original exhibit, panel 2 of 2; read from top to bottom.

5,000; 4-inch products are holding at RMB 7,000, with some reaching nearly RMB 8,000. Compared with July 2026, the current (October 2026) increase is not significant. 6-inch prices have risen much more, from RMB 12,000, 15,000 and 18,000 in April–May 2026 all the way to more than RMB 20,000 now. Looking ahead, prices are expected to hold at this level until new capacity is released, after which they may fall. Among them, 2-inch prices are expected to fall back faster than expected as downstream customers switch to larger sizes.

Q8. What are the current market demand and production yield for 6-inch InP substrates?

Although 6-inch prices have risen quickly, actual downstream demand is not large. On the production side, many manufacturers deployed some 6-inch equipment in 2026; for example, Xiandao already has 120 6-inch crystal growth furnaces on site. However, 6-inch production yield is generally low at present. Taking 100 6-inch furnaces as an example, of the 100 crystals produced in one charge, perhaps fewer than 10 can

Original InP expert-call PDF page 10. Six-inch crystal qualification includes downgrading to smaller wafer sizes. Overseas contract pricing and fixed-price LTAs have separate scopes. These values are not AAOI operating guidance. Original exhibit, panel 1 of 2; read from top to bottom.

ultimately be made into 6-inch finished products, with most of the rest downgraded to 3-inch and 4-inch products.

Q9. What is the current price gap between domestic and overseas InP substrate markets, and how are long-term contract orders priced?

Overseas prices are usually 10% to 20% higher than domestic prices, and the premium on spot prices may be higher. This gap refers to order production prices, not spot prices. Long-term contract orders currently generally use a fixed-price, fixed-volume model, with the specific price level closely tied to signing time because market prices keep changing. For example, long-term contracts signed domestically by Quanlei with Yunnan Germanium are usually multi-year contracts, not renewed annually.

Original InP expert-call PDF page 10. Six-inch crystal qualification includes downgrading to smaller wafer sizes. Overseas contract pricing and fixed-price LTAs have separate scopes. These values are not AAOI operating guidance. Original exhibit, panel 2 of 2; read from top to bottom.

Official ELSFP timing and acceptance

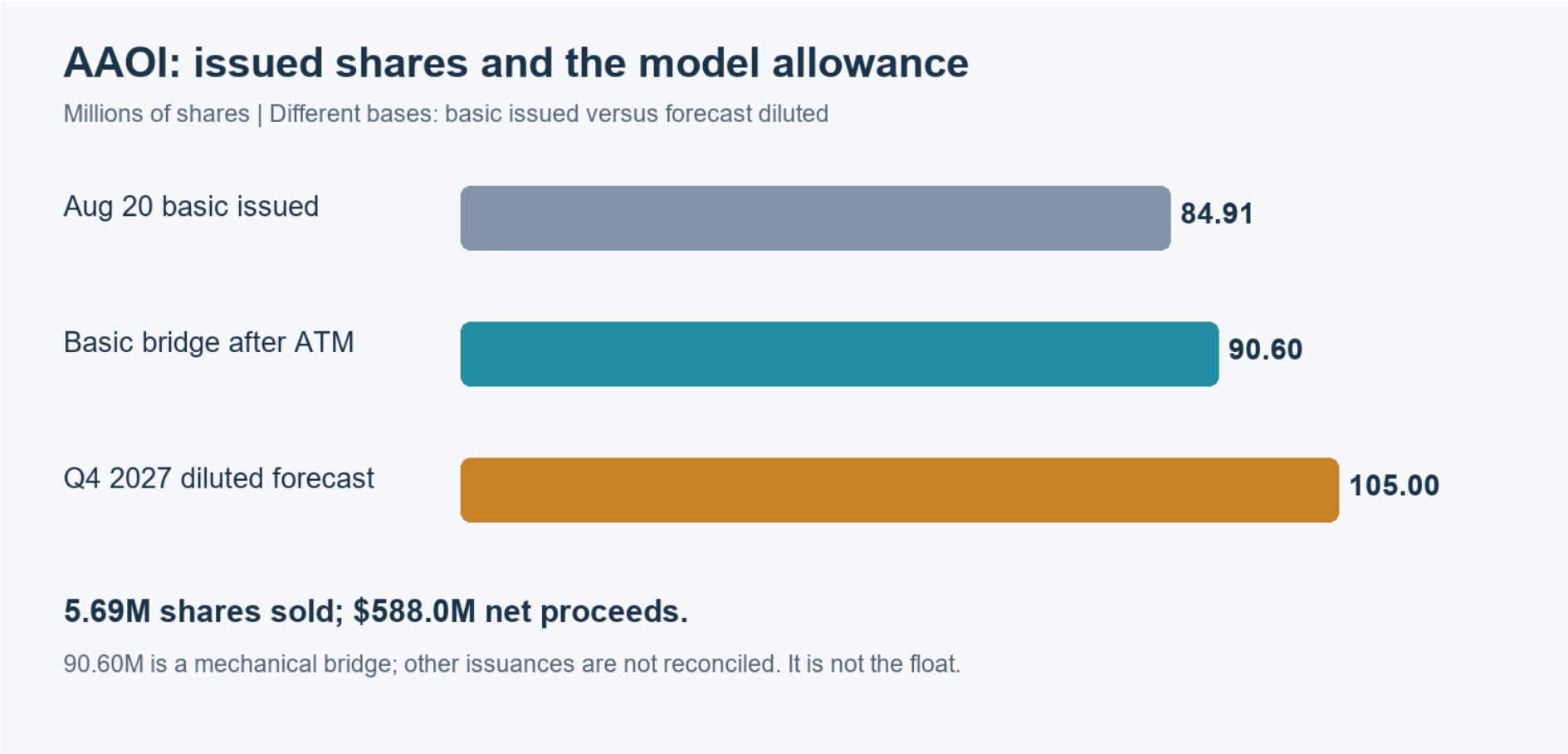
The official Q2 investor deck, listed September 21, 2026 on AAOI's IR page, shows on PDF page 18 nominal 300/400 mW ELSFP module capacity of 5,000 per month in Q4 2026, 50,000 in Q2 2027 and 400,000 in Q4 2027. The prior artifact's comparison against older public-2028 wording is corrected. The current model's nominal end-point already matches the newer deck, so it receives no second capacity addition. The original deck was read through the web PDF and page image; direct byte download failed, and no local original-file hash is claimed.

Capacity milestones do not establish acceptance of the full modeled saleable output. Customer qualifications, recognized orders, high-power die performance, wavelength/linewidth, packaging, lifetime and unit cost remain the evidence tests. The source's possible 200,000 AWS NPO units cannot be inserted as additional ELSFP shipments or used to cap all customer programs. Current all-customer ELSFP revenue, conversion and product costs remain in the financial chapter.

Capital, funding and dilution

The October 6 instrument reconciliation remains the current capital anchor. The new output-delay case changes operating profit; the period-specific economic-share schedule is retained. The active Q4 2027 valuation uses **105 million shares** and **20x forward-four-quarter operating profit**. It uses no independent cash addback or extra Base multiplier.

The October 5 issuer release reports completion of the $600 million ATM announced August 21: **5,694,845 shares** sold and **$587,999,986 net proceeds**, rounded to whole dollars. The latest exact companywide common count located was **84,906,289 as of August 20**. Adding the completed offering once gives **90,601,134**, a mechanical bridge that excludes other net issuance after August 20. The October 5 Form 144 repeats an older August 3 share count and supplies no updated issuer-wide count. Neither count is an estimate of free float. [AS1-AS3, AS12]



Capital bridge: completed ATM shares remain within the existing total-share forecast. The instrument table below supplies the October 6 reconciliation; this original exhibit is not a current legal-share or float estimate.

Capital item	Shares or USD, whole units	Treatment
August 20 issued common	84,906,289 shares	Dated legal common anchor
Completed ATM	5,694,845 shares	Added once
Mechanical common bridge	90,601,134 shares	Other intervening issuance remains unverified
RSUs issuable on vesting	1,090,055 shares	Excluded from August common; avoid duplicating June reserve
PSUs disclosed as issuable	629,463 shares	Target versus maximum payout basis not expressly labeled

Capital item	Shares or USD, whole units	Treatment
Remaining 2030 notes, ordinary physical conversion	About 2,884,318 shares	Principal-based conversion; cash or mixed settlement remains possible
Amazon maximum warrant reservation	7,945,399 shares	Includes initial 1,324,233; full reservation is not immediate issued common
Disclosed gross rights envelope	About 103,150,369 shares	Envelope rather than GAAP diluted weighted average
PSUs at twice disclosed amount	About 103,779,832 total shares	Stress envelope, without another ATM addition
Q4 2027 economic-share allowance	105,000,000 shares	Retained total forecast
Remaining stress reserve	About 1,220,168 shares	Before future discretionary grants or new financing

Issued shares and reservations follow the same instruments once. August common already incorporates the prospectus's 332,428 earlier ATM shares and 188,000 RSU vesting shares, along with prior convertible settlements: 1,487,874 shares in December 2024, 239,404 shares for the last old 2026 notes in July 2025 and 1,731 shares from a small Q2 2026 conversion. August RSU and PSU reservations are excluded from the common anchor. Settlement transfers the corresponding reserve into common. June awards, historical grants, plan availability and outstanding August awards are not separate quantities to stack. The proxy identifies RSU/PSU rights with zero exercise price; no additional outstanding employee-option increment was identified. [AS1, AS2, AS7]

Convertible debt. About $125 million remaining 2030 principal drives the conversion inventory. The approximately $129 million carrying value and $445 million fair value supply accounting and valuation measures, respectively. They do not replace conversion principal. The issuer may settle with cash, shares or both. Early conversion before October 15, 2029 is subject to contractual conditions; issuer redemption from January 15, 2027 also has a specified price test. The maximum special-event make-whole adds about 793,161 shares above ordinary conversion. Event/date/price conditions control this maximum, so it is kept as a stress. The existing target price cannot stand in for an observed market-price test. Old 2026 notes are retired. [AS1, AS5]

Amazon warrants. Full vesting requires $4 billion of qualifying gross payments over the ten-year agreement. The original reserve includes 1,324,233 initially vested shares and up to 6,621,166 purchase-dependent shares. Intermediate purchase thresholds are redacted. The new analyst excerpt's $400 million description cannot replace the original agreement's full-vesting denominator or create a second customer order. The small exercise-price discrepancy between the executed exhibit and later filing remains unresolved; original source precision and contractual terms govern. [AS1, AS4]

H1 2026 warrant-associated revenue was about $85 million against $189 million company datacenter revenue, a reference share of about 45%. The retained timing diagnostic allocates 25%, 45% or 65% of the existing transceiver pool to Amazon, with subjective weights of 25%, 50% and 25%. CATV, Other and ELSFP are excluded. Starting receipt proxies combine FY2025 associated revenue and H1 2026 associated revenue; later cash receipts lag sales by one quarter. A linear vesting diagnostic scales the conditional warrant pool to a $4 billion cumulative-payment proxy, subject to the maximum reservation. Revenue differs from cash paid and the actual intermediate legal steps are unavailable. This remains a diagnostic rather than an active second dilution model. [AS1, AS4, AS6]

Amazon allocation of modeled transceivers	Subjective weight	October 6 diagnostic: Q4 2027 gross warrant proxy
25%	25%	About 3,178,082 shares
45%	50%	About 4,505,035 shares
65%	25%	About 5,831,987 shares

These October 6 payment and vesting diagnostics predate the new 800G delay case. They are retained as dated capital evidence, rather than recast as fresh customer-payment forecasts. The new risk could slow receipts; the unchanged 105 million share allowance already includes the full warrant reservation. The central diagnostic is below that full 7,945,399 share reservation. The disclosed noncash vested-warrant accounting item is a dollar value and supplies no additional share count to add. Contra-revenue from the customer warrant remains separate from dilution; gross versus net product ASP must be reconciled before a further revenue deduction. Cashless warrant delivery also differs from the full gross reservation and depends on the contractual market-price averaging period.

Funding tail. A retained research stress assigns 30% probability to a further $600 million gross equity program by Q4 2027 and 70% to no such program. It remains a financing diagnostic, without a second active share increment. At issue prices of $80, $100 or $150, the offering would create about 8 million, 6 million or 4 million shares and around $588 million cash after a 2% selling cost. A future discretionary award allowance of about 1 million shares, bounded from 500,000 to 1,500,000, is also a research reserve. Full warrant delivery, maximum make-whole conversion and further financing together can exceed the 105 million forecast. The new factory cash-flow statement and actual issuance terms would determine any future change. [AS1-AS10]

Near-term dilution limits remain visible. Q3 2026's 92,800,000 shares were dated guidance before the completed ATM. Unknown daily offering dates prevent an exact quarterly weighted-average adjustment. Q4 2026's retained 96 million forecast is also an economic forecast; adding mechanical common, ordinary conversion, disclosed awards and only Amazon's initial tranche already gives approximately 96,529,203 gross shares. Treasury-stock effects, withholding, settlement methods, vesting and timing determine actual EPS dilution. Historical GAAP Q2 shares were loss-period basic shares; adjusted diluted shares use a different denominator. No new numerical claim of fully reconciled near-term reported EPS is made.

Cash and debt. June cash was approximately $500 million, with about $9 million restricted cash. Subsequent ATM proceeds are financing cash, and a current October balance requires intervening operating and investing flows. H1 operating cash use was about $74 million, cash PP&E purchases $335 million and equipment/other prepayments $290 million. Deposits applied to later equipment are deducted once. A September property purchase used about $27 million. Factory leases, working capital and remaining installations also consume liquidity. Conversion provides no second cash receipt; legal debt and its coupon disappear on settlement, with any cash settlement reducing cash. The unchanged OP valuation has no separate net-cash addback.

An updated issuer-wide legal common count, Q3 weighted dilution, actual Amazon vesting/exercise, redacted thresholds, PSU payout basis and any further funding remain specific watch items. The existing reserve inventory supports retaining the current share schedule while those items are monitored.

Capital sources

- **AS1** Q2 2026 Form 10-Q, filed August 6, 2026. original HTML and extracted text saved.
- **AS2** August 21, 2026 ATM prospectus, common and award counts as of August 20. original HTML and extracted text saved.
- **AS3** October 5, 2026 issuer ATM completion release. original HTML and extracted text saved.
- **AS4** March 13, 2025 Amazon warrant, original executed exhibit. original HTML and extracted text saved.

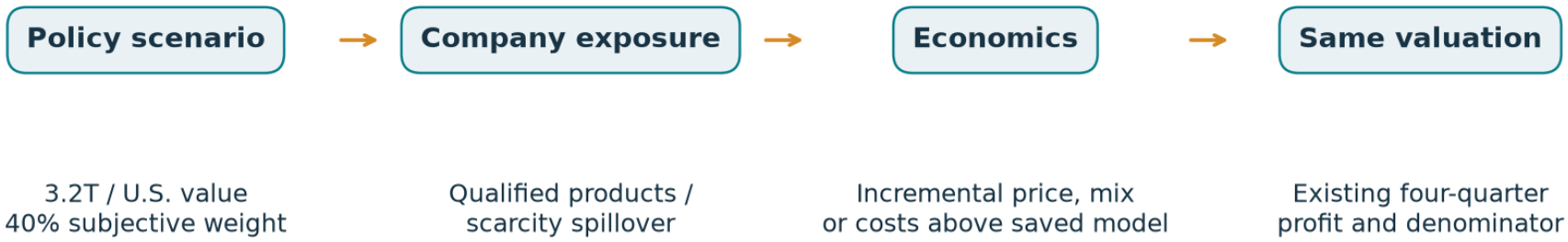
- **AS5** December 23, 2024 indenture for the 3% notes due 2030. original HTML and extracted text saved.
- **AS6** FY2025 Form 10-K, filed February 26, 2026. original HTML and extracted text saved.
- **AS7** April 24, 2026 definitive proxy, equity plans and award statistics. original HTML and extracted text saved.
- **AS8** September 10, 2026 Form 8-K: September 4 property purchase. original HTML and extracted text saved.
- **AS9** September 1, 2026 Form 8-K: August 31 lease agreements. original HTML and extracted text saved.
- **AS10** September 15, 2026 Form 8-K: September 10 Ningbo lease. original HTML and extracted text saved.
- **AS11** August 6, 2026 Q2 results and Q3 guidance. primary page reviewed through web; no local original captured.
- **AS12** October 5, 2026 Form 144, Joshua Yeh. primary page reviewed through web; no local original captured.
- **AS13** September 17, 2026 Form 4, Richard Black. primary page reviewed through web; no local original captured.
- **AS14** September 10, 2026 Form 4, Fred Chang. primary page reviewed through web; no local original captured.
- **AS15** September 10, 2026 Form 4, William Murry. primary page reviewed through web; no local original captured.
- **AS16** June 5, 2026 Form 8-K: June 4 approval of the 2026 equity plan. primary page reviewed through web; no local original captured.
- **AS17** 2026 equity plan Form S-8, 2.5M new shares registered. primary page reviewed through web; no local original captured.

Retained policy hypothesis and production-slot budget

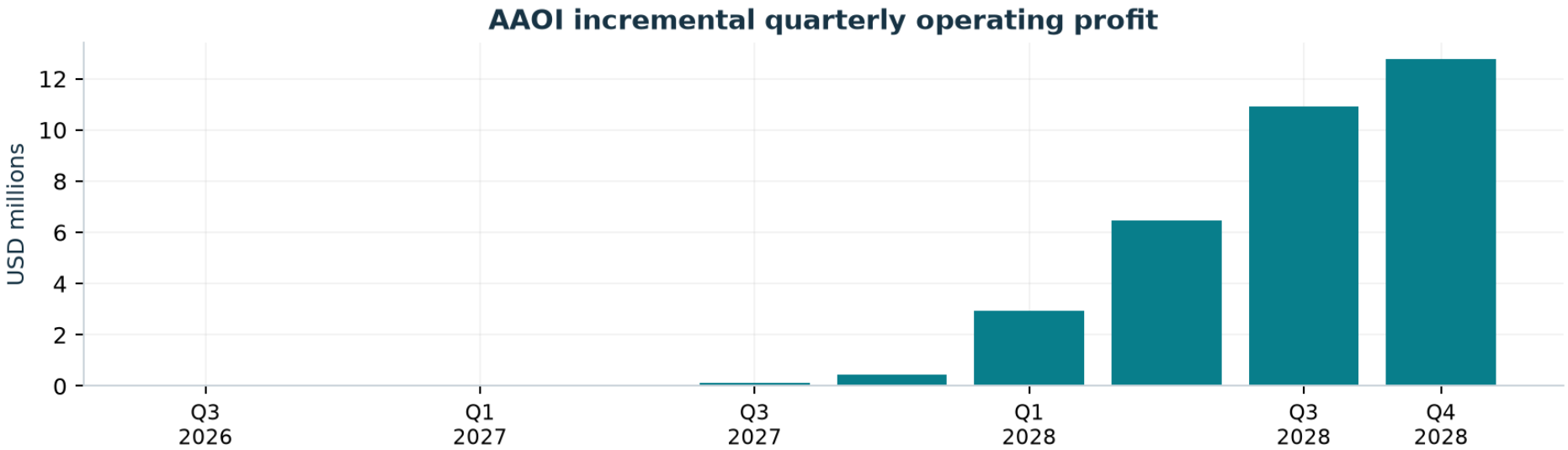
Serenity's October 1 post attributes Washington meetings to Morgan Stanley and discusses a possible 3.2T-focused restriction and a potential 65% US-value route. The final legal definition and original bank note were not recovered. The separate federal-systems bill has its own scope and transition; it does not confirm the exact FCC scenario. Serenity receives the owner's highest source weighting, while the attributed bank and relay remain one underlying chain. Serenity source | foundry follow-up | separate Senate proposal

Retained policy probability: 40%. Conditional AAOI commercialization: 30%; joint weight: 12%. The hypothesis substitutes a bounded amount of existing 1.6T allocation with 3.2T output during 2028. Four 3.2T modules consume five reference 1.6T production slots, and unit COGS is 50% higher. The ASP proxy is $1,200 per module. No factory or free capacity is added; foregone 1.6T sales and GP are deducted. Conditional slot allocation is 5%, 10%, 15% and 15% through the four 2028 quarters.

An external ELSFP pricing branch applies a 5% conditional premium to 50% relevant qualifying and 60% repricable exposure, weighted at the same 40% policy probability. Incremental COGS and opex each consume 5% of incremental price revenue. It uses separate external sales and receives no additional 30% transceiver commercialization multiplier. The pricing phase is zero before Q3 2027, then 25%, 50%, 75% and 100% from Q2 2028. Those are research timing assumptions. No second price-phase multiplier is applied to the explicit 3.2T slot schedule.



Retained policy transmission diagram. Eligibility, product scope, qualified capacity and repricing rights remain uncertain; the current 800G execution case is separately modeled.



Retained incremental policy operating-profit bridge. These policy dollars are preserved; the current 800G risk changes accepted output separately.

Quarter	Policy revenue, USD thousands	Incremental COGS, USD thousands	Incremental opex, USD thousands	Policy OP, USD thousands
Q3 2026	0	0	0	0
Q4 2026	0	0	0	0
Q1 2027	0	0	0	0
Q2 2027	0	0	0	0
Q3 2027	130	6	6	117
Q4 2027	490	24	24	441
Q1 2028	3,875	750	194	2,931
Q2 2028	8,401	1,510	420	6,471
Q3 2028	13,917	2,293	696	10,927
Q4 2028	15,894	2,341	795	12,758

The current 800G case does not alter these policy increments. Eligibility remains an exposure proxy, not an origin certificate. U.S. corporate domicile alone does not establish a product's production origin. Final rule scope, price rights, higher process cost or lower qualified output can reduce the conditional benefit. A higher policy probability requires new evidence beyond reposts of the same chain.

Business research and source evidence

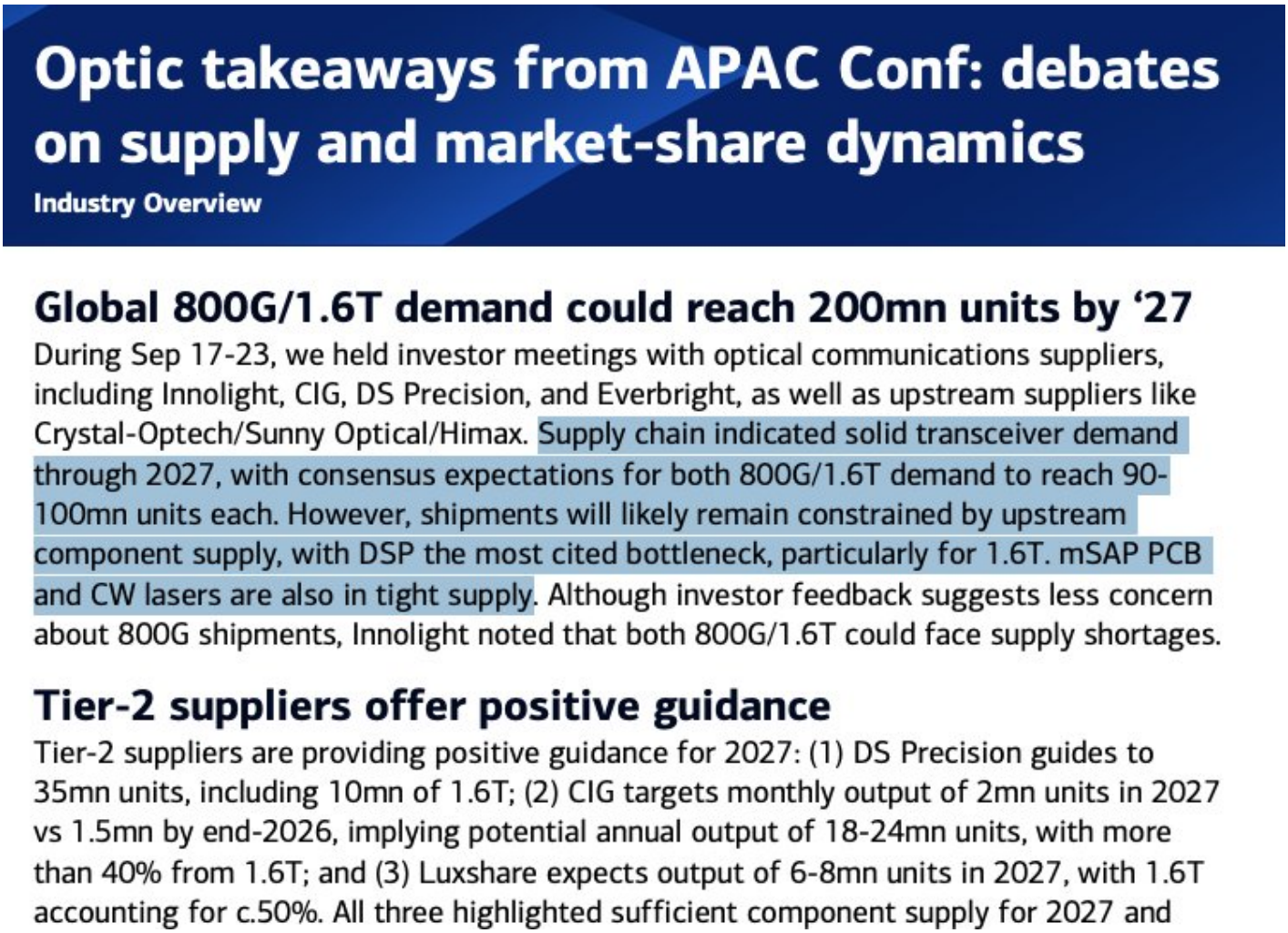
The following sets out the company thesis, historical results, guidance, capacity, program probabilities, risks and original source images. Source dates remain visible. Operating assumptions below feed the complete current financial tables, including the explicit probability-weighted policy contribution.

Inherited breadcrumb inputs and source images

These probabilities are explicit investment judgments under Framework 8.0, not measured statistical frequencies or claims of confirmed orders. Unconfirmed but credible evidence can affect the model when its company-specific transmission is quantified.

Higher800G/1.6T factory utilization

Inherited probability: 30% extra five-point utilization case. Model demand pulling more output through the existing high-speed capacity. Retain the current lower ASP path, shared capacity allocation and 1.6T timing. **Operating bridge:** Conditional 800G and 1.6T utilization rises 5 percentage points from Q1 2027, capped at 95%. Apply probability 30% only to incremental units and their existing product contribution margins. **Limits and counter-evidence:** Both speed lines retain the same allocated capacity: no double use of the factory. DSP/laser shortages can prevent this uplift. Annual GS TAM remains unchanged. Model ends Q4 2028; later price targets stay unavailable.



Optic takeaways from APAC Conf: debates on supply and market-share dynamics

Industry Overview

Global 800G/1.6T demand could reach 200mn units by '27

During Sep 17-23, we held investor meetings with optical communications suppliers, including Innolight, CIG, DS Precision, and Everbright, as well as upstream suppliers like Crystal-Optech/Sunny Optical/Himax. Supply chain indicated solid transceiver demand through 2027, with consensus expectations for both 800G/1.6T demand to reach 90-100mn units each. However, shipments will likely remain constrained by upstream component supply, with DSP the most cited bottleneck, particularly for 1.6T. mSAP PCB and CW lasers are also in tight supply. Although investor feedback suggests less concern about 800G shipments, Innolight noted that both 800G/1.6T could face supply shortages.

Tier-2 suppliers offer positive guidance

Tier-2 suppliers are providing positive guidance for 2027: (1) DS Precision guides to 35mn units, including 10mn of 1.6T; (2) CIG targets monthly output of 2mn units in 2027 vs 1.5mn by end-2026, implying potential annual output of 18-24mn units, with more than 40% from 1.6T; and (3) Luxshare expects output of 6-8mn units in 2027, with 1.6T accounting for c.50%. All three highlighted sufficient component supply for 2027 and

User-supplied evidence for Higher800G/1.6T factory utilization. The quoted forecast or estimate retains the source status described above. Original exhibit, panel 1 of 2; read from top to bottom.

said they had secured DSPs for 1.6T transceivers. However, investor feedback indicates greater confidence in tier-1 suppliers' positioning and a preference to discount tier-2 vendors' guidance because of concerns about DSP supply. Our checks suggest potential DSP volume of 60-70mn units in 2027, broadly in line with most investors' views.

NPO offers potential upside into 2028

All transceiver makers indicated they are working on 3.2T/6.4T NPO projects. Innolight noted that, in a bull case, the market could reach 30-50mn OE units in 2028, vs current expectations of 2-5mn units in 2027 and more than 20mn in 2028. Nvidia's NPO project remains in the sampling stage, and more concrete orders could emerge toward year-end or in early 2027. On design, we see both integrated-laser and ELS approaches, while the supply chain is seeking to use more 100-150mw CW lasers even in ELS solutions.

Passive: more smartphone optic suppliers enter market

More smartphone optical suppliers are expanding into passive components, initially focusing on FAU-related for transceivers with a goal of entering CPO. Crystal-Optech noted that smartphone optical suppliers benefit from greater coating-equipment capacity as equipment lead times have lengthened. However, contributions are unlikely before 2027, as testing/certification typically take a year. Sunny said it is in sampling stage and engaging more closely with domestic customers. On CPO, Himax highlighted its close partnership with FOCI and is working on a 3rd-gen design. Himax noted that 2027 will remain an engineering-run stage but looks positive on future growth.

User-supplied evidence for Higher800G/1.6T factory utilization. The quoted forecast or estimate retains the source status described above. Original exhibit, panel 2 of 2; read from top to bottom.

Sources: APAC.

Transmission field	Inherited treatment
Probability	30% extra five-point utilization case
Recognition	Conditional 800G and 1.6T utilization rises 5 percentage points from Q1 2027, capped at 95%. Apply probability 30% only to incremental units and their existing product contribution margins.
Overlap / uncertainty	Both speed lines retain the same allocated capacity: no double use of the factory. DSP/laser shortages can prevent this uplift. Annual GS TAM remains unchanged. Model ends Q4 2028; later price targets stay unavailable.
Raise probability	Timely qualifying milestones, comparable revenue scope, stronger customer allocation and observed unit economics.
Lower probability	Delay, customer substitution, weaker realized economics, overlap or evidence contradicting the stated source scope.

1. Investment thesis and supercycle position

AAOI provides an investable exposure to the shift toward higher-speed optical modules through its US manufacturing footprint, internal laser capability, and flexible 800G/1.6T assembly lines. The investment thesis requires qualified output and a richer product mix to convert a large market into company revenue. Qualified manufacturing and customer acceptance determine profit capture, while competing suppliers constrain long-term module margins. The operative S-curve is an 800G ramp followed by a faster 1.6T mix shift. External light-source modules for CPO provide another revenue vector, subject to customer qualification, high-power laser capacity and packaging yield. A laser consumed inside an AAOI transceiver is an input cost; it is not a second external revenue sale. The model pools nominal high-speed capacity and retains its reference mix allocation. At Q4 2027, expected 800G conversion is about 90% after the current delay case; 1.6T conversion remains about 92%. The reference capacity allocation shifts toward 1.6T through 2028. Actual expected unit mix also reflects the separate 800G delay. It holds total high-speed capacity at 930,000 units per month during 2028. That is a finite modeling assumption, not a company statement that expansion stops. Pricing declines eventually outrun mix improvement at that capacity level. The 3.2T industry ramp belongs in the active research horizon. AAOI-specific qualification, orders, testing throughput and a production schedule are not established in the materials reviewed, the reference had no 3.2T revenue. The current model assigns a 12% joint probability to a bounded 2028 substitution of existing 1.6T slots, with explicit unit-cost and throughput proxies. Higher market volume alone does not grant AAOI a constant market share or remove component bottlenecks.

TAM expansion lens

Phase	Application and mechanism	Framework treatment	Included company revenue
Phase 1	AI training/inference clusters and hyperscale interconnects; more deployed optical links and greater bandwidth per link	100% thematic weight; company execution still requires evidence	Modeled 800G/1.6T modules and the separately identified ELSFP vector
Phase 2	Wider cloud, edge and physical-AI deployments requiring high-speed optical connectivity	50% thematic weight; assess addressable products and customers	No separately added wallet; avoid overlap with datacenter forecasts
Phase 3	Further distributed-compute architectures and new photonics applications	20% thematic weight; no assumed design wins	Unmodeled

These framework phase weights are research conventions, not probabilities applied to the revenue table. They do not justify adding the same optical demand twice. AAOI's expansion score reflects a widening application set tempered by competition and the need to qualify each product generation.

Engineering: qualification has several separate gates

The March 1 Irrational Analysis article is useful for understanding why an optical module can pass device reliability testing and still wait for customer acceptance. Its reported sequence is reliability and manufacturing-line qualification, followed by interoperability work across the customer's switching platforms. The author proposes a transmitter-equalization coefficient mapping problem as the remaining firmware issue. That is a technically plausible explanation, not an AAOI-confirmed root cause. A subsequent shipment does not retrospectively prove which register, vendor or training algorithm caused the delay. [E-AAOI1]

Michael Edward Genovese: Great. Thanks very much. So it sounds like instead of having a steady kind of ramp, this 800G we are expecting to come in with really big numbers starting 2Q this year and then, you know, huge numbers next year—by 2Q next year. I guess for the ramp in 2Q this year, could you just go over some of the milestones, maybe talk about the issue on the firmware, but also the ongoing qualification milestones that you have to hit to kind of have that ramp in 2Q?

Stefan Murry: Right. So as you mentioned, in order for the ramp to start in earnest, our 800G products have to be interoperable with all the different platforms that are out there at this particular customer. There is a lot of them. So the firmware has to be modified to work with all those different platforms. Hardware-wise, everything is fine. No problem with that. Firmware is good on most of the platforms. We just have to make some tweaks to get it to work across all of the different platforms that they have. And that is really—the customer and us have agreed that should be done in the middle of next month.

So a couple of weeks, three weeks from now, something like that. And then that is basically the last hurdle to kind of unleash the ramp. As we talked about, we have already started manufacturing products for that ramp. So from a manufacturing standpoint, we are gearing up. Just to touch on the beginning part of that question, too, about kind of nonlinearity of the ramp, that is because what you are seeing is our production capacity coming online. Right? It is not gated by demand, it is gated by our ability to produce, and that does not come on in a linear fashion. Right? You build a production line and you get a step function, not a smooth ramp.

Thompson Lin: Yes. Let me emphasize. Because every customer has so many different switch, transceiver supplier, so many different kind of ASIC. And, you know, when a customer adds more switch, this will change the firmware. We were supposed to get a green light to ship out already in December. The delay is not our problem or whatever. It is because, you know, how come build AI—you know, the whole system is much more complicated than before. That is why it takes much longer. And right now, I think we feel very comfortable. And right now, let me say, we have got almost two years of loading forecast from more than one customer. Let me say that, okay? For 800G.

Engineering exhibit aaoi-I07. Saved source: AAOI Probably Worth Chasing, 2026-03-01. Attribution and scope are discussed below.

Historical management-call excerpt reproduced in the March 1 article. Firmware and platform interoperability were management's stated focus at the time. The precise TxFIR diagnosis belongs to the author; it is not a company disclosure. Equalization compensates for an electrical channel's loss and distortion. The article's coefficient examples explain why host/module settings and firmware can prevent a working optical design from passing a particular customer's tests. Its IEEE 802.3ck tables concern electrical interfaces based on 100 Gb/s signaling; they are not a complete 800G optical-module specification. The illustrated seven-bit register mapping is a teaching example, not an AAOI register dump or measured failure. The correct research question is whether the specific customer platform has passed interoperability and acceptance at the required error rate, temperature and operating conditions. [E-AAOI1, E-AAOI3] The historical uncertainty has a dated outcome: AAOI's May 7 release reports its first volume 800G shipment to a large hyperscaler during Q1 2026. Its March 23 order release had also described more than $53M of 800G orders for delivery in Q2 through mid-Q3 following qualification. Those events precede the current Q2 actuals and Q3/Q4 ramp assumptions. They support the existing production path; they are not newly available capacity on October 6. [E-AAOI2, E-AAOI4]

Gate	Economic transmission	What would change the current model
Laser/device reliability	Burn-in, lifetime and thermal failures can raise scrap, screening cost and field returns	Product-specific qualified failure data and associated cost or shipment impact
Manufacturing line	Installed tools become repeatable good output	Accepted yield, test time and qualified monthly shipments, rather than floor area
Host interoperability	A good module may still fail link training or firmware compatibility on one platform	Dated customer/platform acceptance or a new quantified delay
Customer acceptance and delivery	Finished inventory becomes recognized sales	Actual shipped units, realized ASP and acceptance terms

Probability and overlap decision. The historical first volume shipment is a reported fact, recorded at 100% event status. The model retains the separate **30% probability of an extra five percentage points of 800G/1.6T utilization from Q1 2027, capped at 95%**, with existing ASPs, shared capacity and incremental contribution margins. The old firmware-resolution narrative overlaps that already incorporated commercial ramp completely; incremental units, revenue, COGS, opex and OP from this article are all zero. This is a date and overlap decision, not an assumption that unconfirmed engineering signals have no value. No additional Base point is awarded for the same qualification progress. For a future delay, the bounded exposure is the affected customer's product and acceptance quarter. Revenue at risk equals delayed accepted units times that product's ASP; OP impact must deduct avoidable production costs while retaining idle-factory, rework and customer-support expense. Moving all company revenue, or treating delayed units as both lost forever and reshipped later, would overstate the effect. ELSFP remains a separate qualification problem: successful 800G firmware work does not validate high-power laser yield or the model's earlier ELSFP capacity timing. **Engineering sources, reviewed October 6, 2026:** E-AAOI1: Irrational Analysis, March 1, saved full body and all 16 image elements reviewed; E-AAOI2: AAOI Q1 results, May 7; E-AAOI3: IEEE 802.3ck task force and scope; E-AAOI4: AAOI March 23 order release.

2. Actual financial anchor and company guidance

The actual anchor is Q2 2026. Company figures below are distinguished from the quarterly model. [S1, S2]

Q2 2026 actual metric	Value
Revenue	about $192 million
Data center	about $108 million

Q2 2026 actual metric	Value
CATV	about $81 million
Other revenue	about $4 million
Non-GAAP gross profit	about $57 million
Non-GAAP gross margin	30%
Non-GAAP operating expenses	Approximately about $68 million
Non-GAAP operating loss	Approximately about $(10) million
Non-GAAP EPS	6 cents
GAAP net loss	Approximately about $(23) million

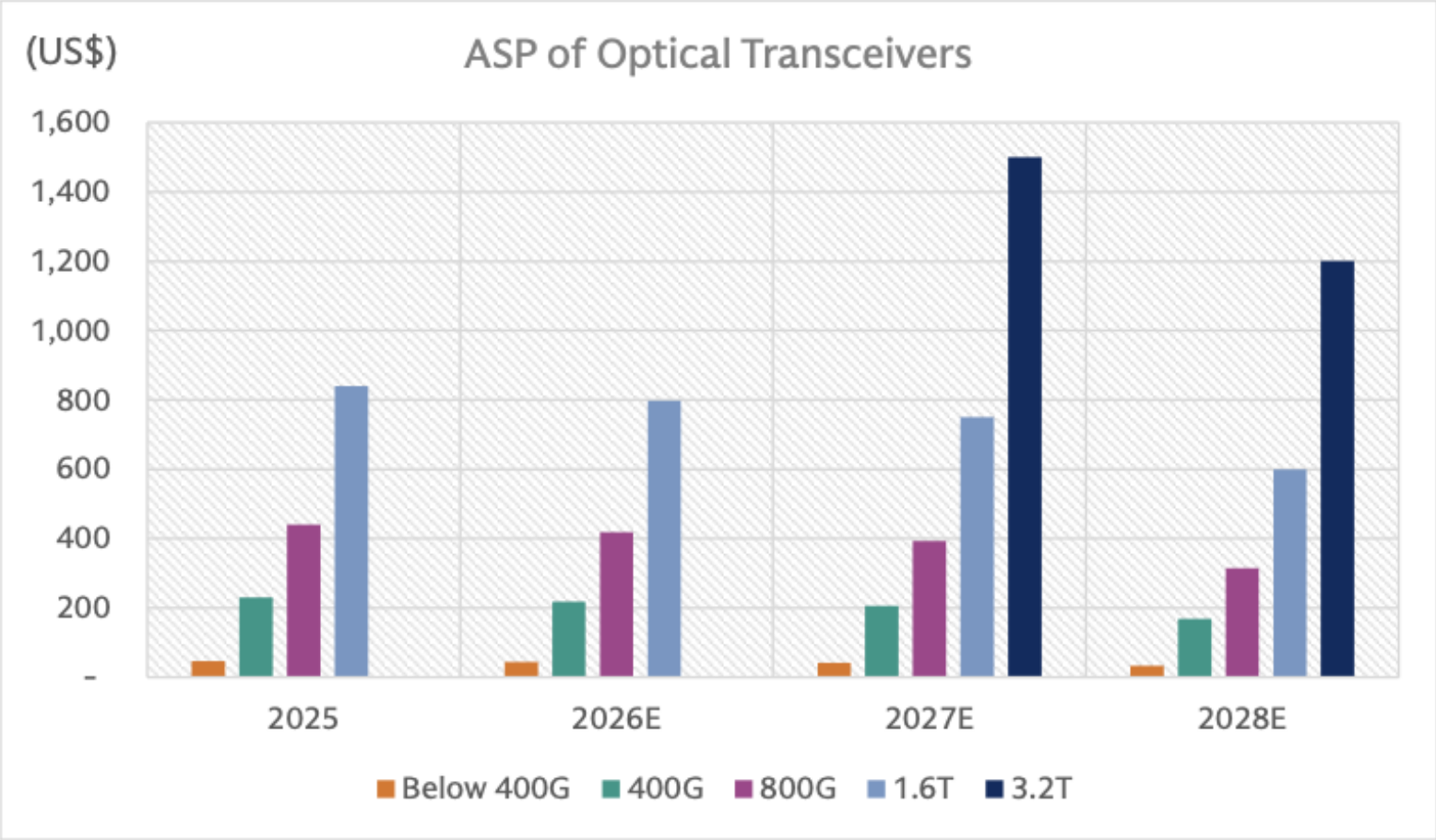
Positive adjusted EPS in that quarter does not establish positive operating profit. The valuation below uses modeled operating profit, not EPS, subsidy income or tax benefits. **Formal Q3 2026 guidance:** revenue $255-$290 million, non-GAAP gross margin about 29%-31%, and approximately 92,800,000 diluted shares. [S1] The management call supplies additional operating-expense, product-ramp and capacity context. Those statements are dated management expectations rather than reported subsequent results. [S2] The prior company artifact records management's expectation of more than $500M revenue in Q4 2026, a $70-80M 1.6T contribution, Q1 2027 1.6T revenue at least doubling, and a mid-2027 datacenter-transceiver milestone of roughly $471M monthly. It also records a $200M 1.6T purchase order and customer demand above production capacity. These are company-specific demand anchors, not additional amounts to add to the quarterly revenue schedule. [S2, S6] The current Q4 2026 blended gross margin rounds to 33% after the 800G risk and retained factory costs. Product margins, including the retained five-point 1.6T reference premium, are assumptions. The old reference model rounded to 34% blended GM; neither product-level margin path is issuer guidance.

Company capacity milestone	High-speed modules per month	Treatment
End Q1 2026	About 100,000	Reported historical capacity context
At Q2 2026 call	Approaching 200,000	Dated management statement
End 2026	Around / over 650,000	Management forecast endpoint
End 2027	Over 930,000	Management forecast endpoint
During 2028	No incremental high-speed capacity established in this model	Model holds 930,000; expansion remains an open research input

Management describes 800G and 1.6T as sharing manufacturing lines, with different final testing requirements. The model uses that flexibility to change product mix. Pooling nominal assembly capacity does not prove interchangeable final-test throughput, sufficient DSP/TIA availability, laser capacity, or customer qualification. [S2] The inherited company material describes Texas capacity additions, including initial production at the 210,000-square-foot facility in late Q3 2026 and additional facilities expected in early 2027. Monitor actual production acceptance rather than treating construction area as shipped units. The memory-related switch shortage can also suppress legacy 100G shipments even while high-speed optics demand rises. [S2, S6]

3. Optical market baseline and company capture

Exhibit 4: Global Optical transceivers: Assuming ASP decline YoY for like-for-like products



Source: Company data, Goldman Sachs Global Investment Research

Supplied GS Exhibit 4. Module ASP benchmarks; quarterly AAOI pricing retains the original interpolation.

Use Framework 8.0 DB1 for market definitions and source status. The global module/transceiver forecast reported for September 2026 is **$68 billion in 2026, $131 billion in 2027 and $149 billion in 2028**. It is not a company revenue forecast or an additive sum of modules, lasers, substrates and CPO engines. [S3, S4] The public GS summary reports approximately 49M 800G, 71M 1.6T and 23M 3.2T units in 2027. Its 61% increase to the 2027 1.6T-and-above shipment forecast is a revision to the broker's earlier forecast, not a growth factor to multiply into AAOI's units. [S3]

Approximate annual module ASP	2026E	2027E	2028E
800G	$420	$400	$320
1.6T	$800	$750	$600
3.2T	Not shown	$1,500	$1,200

ASPs are rounded visual readings of the user-supplied Goldman Exhibit 4. They are broad industry benchmarks, not disclosed AAOI realized prices or a like-for-like match for every reach, architecture, customer and manufacturing location. The retained quarterly price path was calibrated to the annual benchmarks using the reference shipment weights and adds no general US-production premium. The separate current ELSFP premium applies to its filtered exposure only. Any premium requires a SKU/customer/contract bridge. [S4] The reported Rubin Ultra assumptions are two 1.6T and five 3.2T modules per GPU; these are broker-model assumptions, not a verified AAOI order or NVIDIA production BOM. The SiPh percentages reported in the GS summary refer to speed categories: 60% for 800G, 80% for 1.6T and 80% for 3.2T. They do not automatically determine AAOI's architecture mix or laser requirements. [S3, S4] The supplied CPO shipment revisions are -33% for 2026, +35% for 2027 and +105% for 2028, each relative to the preceding broker shipment forecast. No absolute CPO dollar market or AAOI ELSFP growth rate is inferred from those percentages. ELSFP uses its own capacity, conversion and ASP schedule below. [S4] **2027 company capture cross-check:** about 3,933,000 expected 800G units and 3,753,000 1.6T units. These are model output requirements, not awarded market shares. Global demand forecasts do not eliminate component constraints.

4. Base score and risk deductions

The analytical score is 42/60, rounded to 70%, following the current qualification and cycle review. U.S. manufacturing was already credited; it is not scored again. All portfolio Excel Base values remain protected. The current category table is shown in the analytical Base chapter. Excel Base remains 72 under the owner lock. The Base score is a thesis-quality measure. It does not independently validate the modeled utilization, margins or optionality, and no additional Base multiplier is applied to the operating-profit price target.

5. Eight-vector model and source hierarchy

All ten forecast quarters carry **Model** status, including quarters constrained by management guidance. Actual results retain a separate anchor. Product-level forecasts are not upgraded to HIGH confidence simply because an aggregate model fits a guided range.

Revenue vector	Revenue derivation	Evidence / source	Forecast confidence
10G/40G	Explicit quarterly revenue schedule	S6 model assumption; Q2 company product context	LOW
100G	Capacity K/month × 3 × ASP × conversion / 1,000	S6 inherited unit schedule; S2 legacy-demand context	LOW

Revenue vector	Revenue derivation	Evidence / source	Forecast confidence
200G/400G + AOC	Combined capacity K/month × 3 × ASP × conversion / 1,000	S6 mixed-product schedule; no supported SKU split for repricing solely from 400G chart	LOW
800G	Pooled high-speed capacity × utilization × 800G mix × ASP	S2 company capacity/flexibility; S4 price benchmark; explicit modeled utilization/mix	LOW at product level; near-term total revenue is guidance-constrained
1.6T	Pooled high-speed capacity × utilization × 1.6T mix × ASP	S2 company orders/qualification; S4 price benchmark; explicit modeled utilization/mix	LOW at product level; near-term total revenue is guidance-constrained
ELSFP	Independent capacity K/month × 3 × $400 ASP × conversion / 1,000	S2/S6 capacity and module context; modeled recognition/qualified-output conversion	LOW
CATV	Explicit quarterly revenue schedule	S2 near-term demand context; S6 modeled later run rate	MEDIUM near term; LOW thereafter
Other	Explicit quarterly revenue schedule	S6 residual-business assumption	LOW

The company-level proxy is capacity × qualified saleable output × realized price, supported by company demand milestones and checked against the global speed-specific market. A customer-capex-to-AAOI procurement allocation is not available at sufficient granularity to establish individual customer shares. No such shares are fabricated. Forecast confidence remains limited by that gap.

Quarterly revenue derivation

Each high-speed generation uses its allocated share of nominal capacity multiplied by three months, qualified saleable conversion and the retained ASP. The inherited 30% extra-five-point utilization case remains inside the reference. A separate 25% chance of a one-quarter lag reduces only 800G conversion during Q4 2026 through Q4 2027. The 1.6T path and capacity allocation remain unchanged; constrained 800G slots are not automatically reassigned to 1.6T. The quarterly ASP path is retained. The annual GS price-shape normalization uses the original shipment weights; changed utilization can make active sales-weighted annual ASP differ slightly from the original benchmark. This avoids repricing the model unintentionally. The prior positive utilization case retained its original operating-expense transmission. For the new adverse 800G state, research staff, ramp support and other quarterly opex are retained. Of the COGS attached to delayed units, 85% is avoided and 15% remains as a labeled underabsorption proxy. This generates the displayed lower product margin without adding a second idle-factory penalty.

Guidance and market calibration

Q3 2026 modeled revenue of about $271 million and blended gross margin of 30% fall inside formal guidance. Modeled operating expenses of about $74 million fall inside the range recorded in the management call. This anchors the near-term model but does not validate 2027-2028 product ASPs, conversion or costs. [S1, S2] Q4 2026 revenue is about $469 million after the 800G delay case, with unchanged 1.6T revenue of about $76 million and a rounded 33% blended gross margin. The reference revenue before this risk was about $507 million. The current forecast is below management's earlier more-than-$500-million expectation because the new downside case is weighted into accepted output. Q1 2027 1.6T revenue remains about $321 million; its mix/volume assumptions are separate from a specific issuer dollar guide. Management’s mid-2027 monthly milestones are retained below as dated source evidence. A run-rate milestone and a quarterly average are not identical periods; the current financial chapter supplies the new model comparison.

Monthly transceiver revenue $M	Management mid-2027 milestone
100G/400G	90
800G	217
1.6T	164
Total, excluding ELSFP/CATV/other	471

The product allocation requires separate checks. The 1.6T-heavy mix requires additional qualification, components and final-test capacity. Similar aggregate totals do not prove line-by-line convergence; no artificial product plug is inserted to force agreement.

Rolling forward-four-quarter price target

The price target uses operating profit in the stated quarter plus the next three quarters. The company-specific method remains **20x modeled operating profit**. USD millions / period-specific diluted shares in millions. Only seven complete valuation windows; no extrapolated 2029 earnings. Probability is applied to the specified operating increment once; the price target receives no second probability or Base multiplier.

Annual cross-check and demand scale

Calendar buckets sum the stated quarter labels; fiscal proxy mappings remain as in the company earnings section. Annual totals do not replace rolling valuation windows.

Operating driver and probability transmission

No active 2029-2030 operating projection is introduced. Industry demand does not establish additional AAOI capacity beyond the retained schedule. Whole-company revenue including CATV and ELSFP is not divided by a module-only TAM.

9. 3.2T, CPO and optionality ledger

3.2T research gate: the GS industry forecast places this generation in 2027-2028. The reviewed AAOI product overview identifies 800G, 1.6T and CPO/NPO offerings, but it does not establish a company-specific 3.2T volume ramp. Required inputs are a named product/customer qualification, recognized order or allocation, production start, final-test capacity, unit throughput, ASP, margin and displacement of existing output. [S3, S5] At approximate annual industry ASPs, 100,000 genuinely additional 3.2T modules in a quarter would represent $150M revenue in 2027 or $120M in 2028. If they instead replace 1.6T units one-for-one, the illustrative net revenue increment is $75M or $60M at the annual benchmarks. These arithmetic cross-checks contain no assumed AAOI order, ramp date, throughput equivalence or added profit. No amount from them is included in the current quarterly table. **CPO and high-power laser engineering:** external-light-source performance, wavelength control, laser die area, production yield, reliability qualification and packaging yield must be assessed separately. The earlier company material records 300-400mW development and DWDM yield challenges; it does not validate every shipment in the forward ELSFP schedule. Management's component supply position must also be distinguished from DSP/TIA availability and final assembly. [S2, S6]

Item	Current evidence treatment	Revenue treatment	Required next evidence
800G/1.6T company ramp	Company demand/capacity anchors plus explicit model utilization and mix	Reference capacity/mix plus the inherited 30% utilization case and current 25% 800G delay case	Recognized output, qualification, pricing and product costs
ELSFP	Company product/capacity context; LOW-confidence timing and conversion assumptions	Included once as external module sales	Qualified high-power output, customer acceptance, unit yield and recognized sales
AAOI 3.2T	Company-specific ramp remains LOW confidence	Modeled: 40% policy x 30% conditional conversion = 12%; slot and cost limits in the financial chapter	Product/customer qualification, orders and production schedule
High-speed capacity beyond 930,000 per month	Not established in the active schedule	Unmodeled	Company expansion timeline, qualified equipment and financing
AAOI pricing premium	A filtered external ELSFP premium is a current assumption, not a quote	5% conditional premium on 50% qualifying x 60% repricable ELSFP exposure; 40% weight	Contract or realized SKU/customer pricing
Broader physical-AI and edge markets	Thematic research opportunity	No separately added revenue	Addressable product and non-overlapping demand bridge

Base and modeled conversion have distinct roles. Conversion represents saleable/recognized output; it is not a hidden probability discount. The accepted positive utilization increment remains inside the reference 800G/1.6T vectors; the current adverse case changes only 800G accepted output; the policy model separately identifies the 3.2T substitution vector and subtracts displaced 1.6T sales.

10. Competitive position and cross-book implications

AAOI's US footprint and internal laser capabilities may help delivery, customer qualification and sourcing diversification. They do not by themselves establish a durable module monopoly or a guaranteed ASP premium. Competitive pressure must be evaluated by product reach, architecture, customer allocation, quality and manufacturing cost. An LTA needs a review of price formulas, minimum purchases, allocation rights, resets and cancellation terms. Declining an LTA does not prove prices will rise or that all competitor contracts contain both a price floor and price cap. A signed agreement is not automatically a thesis break; its economics control.

Paired exposure	Shared variable	What to check independently
LITE / COHR / SIVE	High-speed optics and external-light-source demand	Product overlap, qualified laser output, optical power and external sales; avoid treating all laser designs as interchangeable
MRVL / SMTC	DSP, TIA and related component availability	Whether components support AAOI's planned product mix and shipment schedule
AXTI / IQE and substrate/epi suppliers	InP and compound-semiconductor production	Qualified input supply, yield and material intensity; module counts are not wafer starts
TSEM / GFS	SiPh adoption	AAOI-specific architecture and foundry/customer sourcing, rather than industry penetration alone
MU / SNDK / SK hynix / Samsung	Memory constraints and datacenter deployment	Separate positive memory pricing from switch shortages that can delay legacy transceiver demand

These links identify research dependencies. They do not transfer AAOI's revenue forecast, margin assumptions or risk score to another company.

11. Risks and falsification tests

The five risk classes below correspond to the five one-point deductions in the Base table.

1. **Execution and component supply (-1).** The projection requires new capacity to become qualified output, utilization to rise, and product allocation to shift toward 1.6T. Final testing, DSP/TIA supply and laser production can constrain output despite nominal assembly capacity. Counter: material production delay or persistent failure to convert installed capacity into recognized shipments. A Q4 2026 revenue result below $400M would trigger a full thesis review; this is a research threshold, not an instruction to trade.
2. **Competition, concentration and price/cost economics (-1).** Customer concentration and standardized module competition limit pricing power. The GS benchmarks may differ from AAOI's SKU mix; preserving projected margins requires the displayed cost curve. Counter: realized ASP, yield or cost evidence that invalidates the modeled product margins, or customer allocation loss. High-speed price decreases without corresponding cost improvement directly undermine the profit model.
3. **ELSFP performance, yield and timing (-1).** High-power optical performance, wavelength control, external-module qualification and packaging yield may delay recognition. The official Q2 investor deck listed September 21, 2026 shows a Q4 2027 nominal ELSFP milestone of 400,000 modules per month. That corrects the earlier artifact's stale public-2028 timing comparison. Nominal capacity still requires qualified chips, packaging, customer acceptance and recognized orders. The model's conversion schedule is separate from a reported mechanical yield. Counter: an announced delay beyond 2028 or no recognized ELSFP revenue by Q4 2027; revise the vector and all affected four-quarter windows.

4. **Funding and dilution (-1).** Factory expansion and working capital can require equity or other financing. Share counts in the price target path are assumptions after the near-term company guide; operating profit is not free cash flow. Counter: share issuance materially above the period-specific schedule or a funding plan that changes capacity timing. Recompute price targets using the actual diluted-share bridge.
5. **Legacy demand and architecture transition (-1).** Memory-related switch constraints can suppress 100G demand, while a faster architecture transition can displace 800G before AAOI captures the next product generation. Counter: persistent legacy weakness, lost design wins or customer adoption of a product AAOI cannot supply on the modeled schedule.

No sixth risk point is added for the same economics under another label. Supplier agreements, LTAs and geopolitical sourcing developments feed the relevant class above. The current category review supports the explicit 42/60 total.

12. Dated catalysts and counters

Window	Required development / positive evidence	Counter / model response
Q3 2026 results	Revenue within the company guide; confirm 1.6T qualification and recognized shipments	Below-guidance results or qualification delay: replace estimates with actuals and rebuild the ramp
Q4 2026 results	Shipment step toward current modeled $469 million revenue and $76 million 1.6T sales after the probability-weighted 800G delay	Reconcile realized margins, product output and the 33% rounded current GM; the original 34% reference was an assumption
H1 2027	Installed equipment becomes qualified output; demonstrate common-line mix flexibility	Testing, DSP/TIA or laser bottlenecks prevent the utilization/mix progression
Mid-2027 / Q3 2027	Track progress toward management's monthly transceiver milestone and test the modeled product split	Aggregate revenue alone does not validate 1.6T share, ASPs or margins
Q4 2027	930,000/month shared high-speed nameplate; about 90% expected 800G and 92% 1.6T conversion; evidence for ELSFP qualified output	Capacity exists but is not saleable, or ELSFP timing fails the model assumption
2027-2028	Named AAOI 3.2T qualification, order and manufacturing evidence	Reassess the 12% Modeled joint probability, throughput and cost proxies against company-specific progress
Every 2028 quarter	Validate realized ASP and unit-cost declines; assess incremental expansion	Capacity stays fixed while pricing erodes faster than mix improves
Each filing / financing update	Reconcile diluted shares and cash required to fund output	Share count or funding needs exceed the stated schedule

The escalation path is named evidence → qualified design/product → customer order/allocation → recognized shipment. Source credibility can raise research priority before revenue is booked, but timing and unit economics must enter the quarterly model explicitly. Refresh the applicable four-quarter valuation windows whenever actuals, guidance, costs, capacity or shares change.

13. Source register and confidence limits

- **S1 | Company actuals and formal guide:** Applied Optoelectronics Q2 2026 results, August 6, 2026; checked September 20. Primary company release.

- **S2 | Management statements:** AAOI Q2 2026 management call transcript, August 6, 2026. Use attributed management statements; a third-party transcript can contain transcription errors. Company webcast is linked from its events page.
- **S3 | Broker forecast summary:** Public summary of Goldman Sachs's September 7, 2026 optical-transceiver report, checked September 20. The original full Goldman research note was not inspected. This is secondary-source corroboration of the broker forecast, not an independent demand signal.
- **S4 | User-supplied market evidence:** Goldman Exhibit 4 ASP chart and the September 8 market / Rubin Ultra post and ASP / CPO revision post. The chart readings are approximate. The source image is delivered as framework_sources/Goldman_Exhibit4_Optical_ASP_user_supplied.png. Framework 8.0 DB1 defines the scope and limitations.
- **S5 | Company product scope:** AAOI optical-transceiver products, checked September 20. Reviewed materials did not establish a dated AAOI 3.2T volume-production schedule.
- **S6 | User's existing model and artifact:** v3_2_master_portfolio.xlsx, AAOI_Q_Model, and the supplied full AAOI artifact. These supply inherited lower-speed and ELSFP unit schedules, product-margin assumptions, dollar opex and diluted-share assumptions. The source workbook SHA-256 is 387723f831b491f22c17656dd182c3c1a9f7c6697440c2b066f67ca682a3f620. Company-slide values cited inside that model are inherited where the original presentation was not independently retrieved.
- **S7 | Active model decision:** the September 20 user discussion of the GS-price / accelerated-ramp projections and instruction to update the full artifact. The unit mix, utilization, quarterly price interpolation, margin/cost requirements and finite horizon in this document are explicit analyst assumptions authorized for incorporation; authorization does not turn them into disclosed company facts.

Open evidence requirements: AAOI realized ASP by comparable SKU/customer; test-limited 1.6T allocation on shared lines; unit-cost and margin bridge; ELSFP timing/conversion and recognized revenue; additional high-speed capacity after the disclosed milestones; 3.2T orders and qualification; actual Q3 weighted dilution, an updated company-wide legal count, Amazon purchase/vesting detail and future funding against the reconciled total-share allowance. These gaps limit confidence and remain visible in the model rather than being filled with invented inputs.

Current-state controls

This is the current production analysis. The financial model includes the probability-weighted policy contribution; all portfolio Excel Base values remain protected.

Model reconciliation and limitations

The current financial chapter supplies the new forecast tables. Excel Base overrides, valuation methods, share schedules and units remain protected. The next 12-16 months deserve the highest forecast weight; outer-year estimates are substantially less reliable. Revisit probabilities and operating inputs as evidence changes. Official confirmation is not required to value a quantified breadcrumb; source uncertainty remains explicit.

Complete current operating model

The model covers Q3 2026 through Q4 2028. All forward rows are labeled Model; historical results and formal guidance remain separate. Financial figures below are USD millions rounded to whole units. Small displayed differences can arise from rounding; exact vectors, margins, cost bridges and all nine execution sensitivities remain in the linked model.

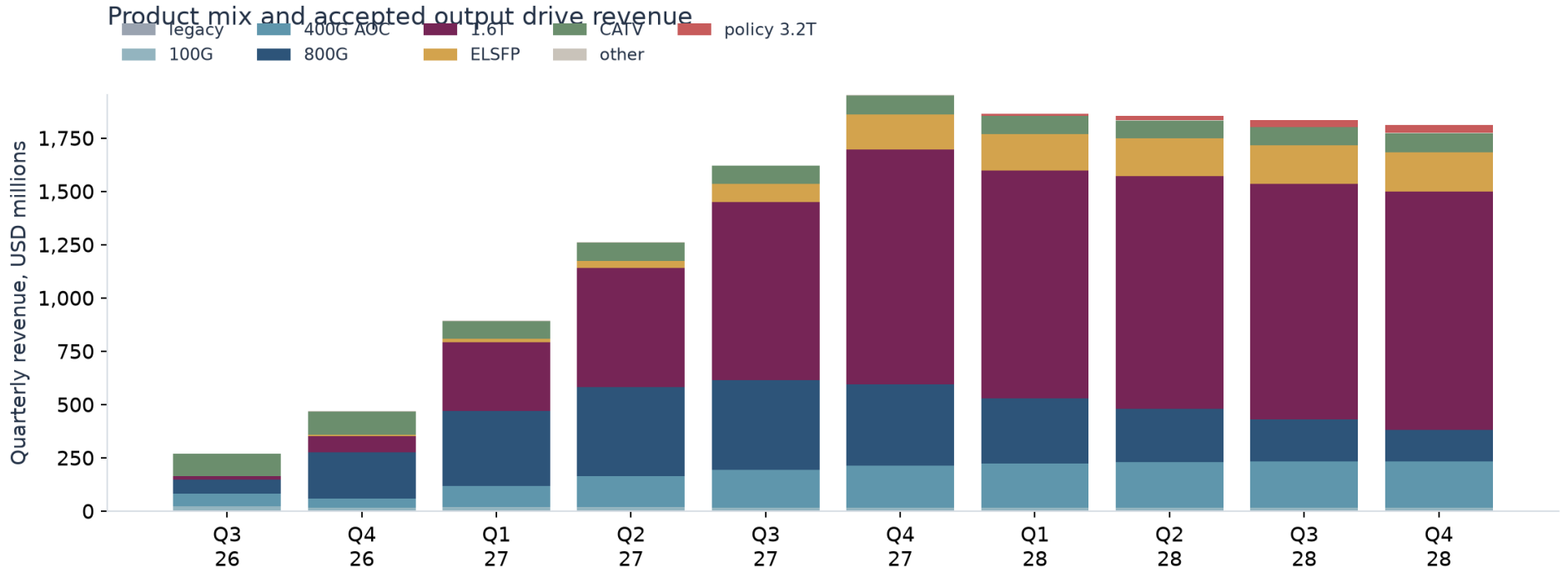


Figure 3. Current quarterly revenue by product. All eight existing vectors remain; policy-linked 3.2T substitutes existing 1.6T production allocation.

Quarter	Revenue	Gross profit	Gross margin	Opex	OP	OP margin
Q3 2026	271	82	30%	74	8	3%
Q4 2026	469	156	33%	82	74	16%
Q1 2027	894	323	36%	110	214	24%
Q2 2027	1,262	484	38%	133	351	28%
Q3 2027	1,622	658	41%	163	495	31%
Q4 2027	1,952	828	42%	195	633	32%
Q1 2028	1,865	805	43%	203	602	32%
Q2 2028	1,855	809	44%	208	601	32%
Q3 2028	1,835	803	44%	210	592	32%
Q4 2028	1,810	788	44%	210	578	32%

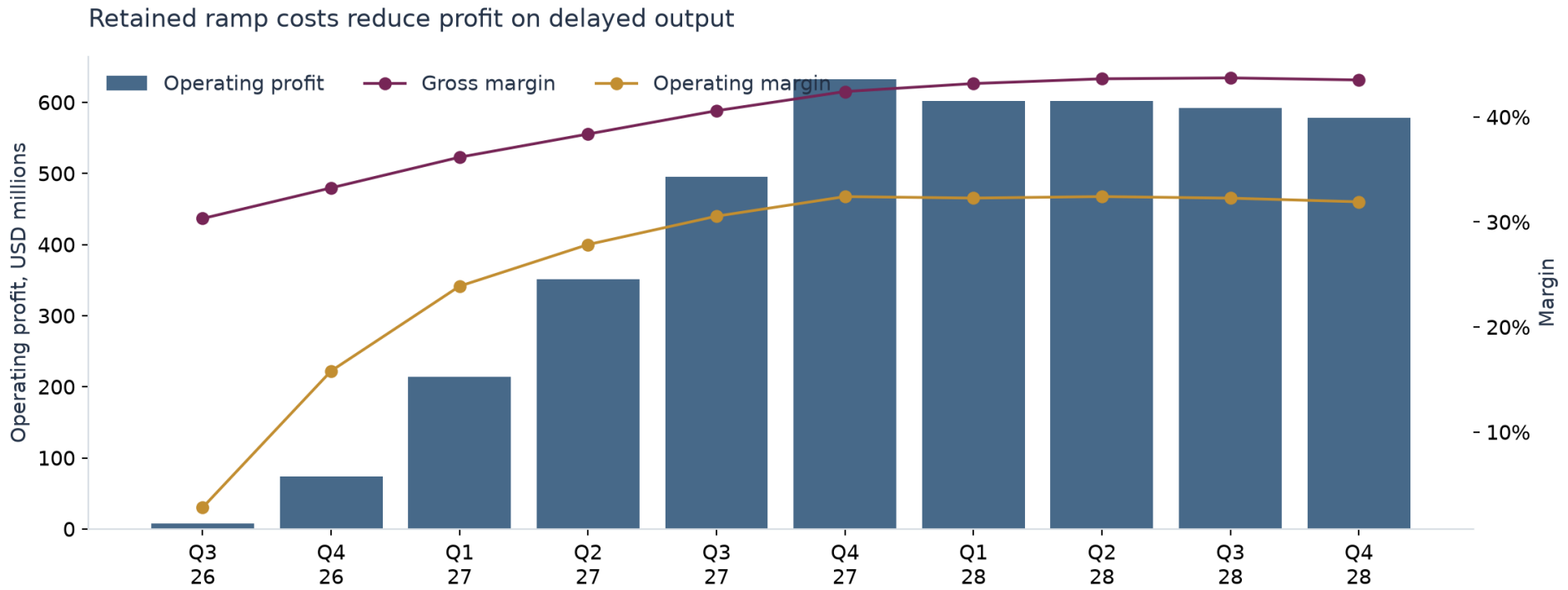


Figure 4. Current operating profit and margins. The 800G adverse state retains opex and a portion of displaced unit COGS; no extra Base multiplier is used.

10G/40G legacy

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	4	0	12%
Q4 2026	3	0	12%
Q1 2027	4	0	12%
Q2 2027	3	0	12%
Q3 2027	3	0	12%
Q4 2027	3	0	12%
Q1 2028	3	0	12%
Q2 2028	3	0	12%
Q3 2028	3	0	12%
Q4 2028	3	0	12%

100G

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	19	4	20%
Q4 2026	12	2	20%
Q1 2027	15	3	21%
Q2 2027	15	3	21%
Q3 2027	14	3	21%
Q4 2027	14	3	21%
Q1 2028	13	3	21%
Q2 2028	13	3	21%
Q3 2028	13	3	21%
Q4 2028	12	2	20%

200G/400G and AOC

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	60	19	31%
Q4 2026	45	15	32%
Q1 2027	100	34	34%
Q2 2027	147	51	34%
Q3 2027	176	61	35%
Q4 2027	198	69	35%
Q1 2028	208	73	35%
Q2 2028	215	75	35%
Q3 2028	217	76	35%
Q4 2028	217	75	35%

800G

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	64	17	26%
Q4 2026	215	67	31%
Q1 2027	351	119	34%
Q2 2027	417	150	36%
Q3 2027	423	160	38%
Q4 2027	379	148	39%
Q1 2028	304	121	40%
Q2 2028	248	99	40%
Q3 2028	197	78	40%
Q4 2028	149	59	39%

1.6T

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	17	5	31%
Q4 2026	76	29	38%
Q1 2027	321	128	40%
Q2 2027	559	232	42%
Q3 2027	832	358	43%
Q4 2027	1,102	487	44%
Q1 2028	1,068	479	45%
Q2 2028	1,091	491	45%
Q3 2028	1,105	494	45%
Q4 2028	1,118	494	44%

External ELSFP

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	0	0	50%
Q4 2026	6	3	50%
Q1 2027	16	8	50%
Q2 2027	33	17	51%
Q3 2027	87	45	52%
Q4 2027	164	87	53%
Q1 2028	171	93	54%
Q2 2028	178	98	55%
Q3 2028	181	101	56%
Q4 2028	183	103	56%

CATV

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	105	37	35%
Q4 2026	110	39	36%
Q1 2027	85	30	36%
Q2 2027	85	31	36%
Q3 2027	85	31	36%
Q4 2027	90	32	36%
Q1 2028	85	31	36%
Q2 2028	85	31	36%
Q3 2028	85	30	36%
Q4 2028	90	32	36%

Other

Quarter	Revenue, USD M	Gross profit, USD M	Gross margin
Q3 2026	2	0	20%
Q4 2026	2	0	20%
Q1 2027	2	0	21%
Q2 2027	2	0	21%
Q3 2027	2	0	21%
Q4 2027	2	0	21%
Q1 2028	2	0	21%
Q2 2028	2	0	21%
Q3 2028	2	0	21%
Q4 2028	2	0	20%

Unit drivers and constrained output

The allocated capacity below is the same reference pool as before this review. Expected conversion is specific to the product generation; only 800G changes under the selected case. Quarterly ASPs are retained, so changing the weighted shipment mix can change the realized annual blended price without repricing a contract.

Quarter	800G allocated K/month	800G ASP, USD	800G expected units K	1.6T allocated K/month	1.6T ASP, USD	1.6T units K
Q3 2026	351	420	152	48	800	21
Q4 2026	562	420	513	88	800	95
Q1 2027	500	425	827	214	812	396
Q2 2027	468	409	1,021	312	781	716
Q3 2027	428	393	1,077	428	750	1,109
Q4 2027	372	377	1,007	558	720	1,532
Q1 2028	326	340	893	604	648	1,659
Q2 2028	279	324	766	651	618	1,787
Q3 2028	232	308	638	698	588	1,915
Q4 2028	186	293	511	744	557	2,042

Quarter	ELSFP nominal K/month	ELSFP conversion	External revenue, USD M	1.6T policy adjustment, USD thousands	3.2T revenue, USD thousands
Q3 2026	2	0%	0	0	0
Q4 2026	5	100%	6	0	0
Q1 2027	27	50%	16	0	0
Q2 2027	50	55%	33	0	-0
Q3 2027	225	32%	87	0	0
Q4 2027	400	34%	164	0	0
Q1 2028	400	36%	171	-6,449	9,558
Q2 2028	400	37%	178	-13,245	20,586
Q3 2028	400	38%	181	-20,248	33,085

Quarter	ELSFP nominal K/month	ELSFP conversion	External revenue, USD M	1.6T policy adjustment, USD thousands	3.2T revenue, USD thousands
Q4 2028	400	38%	183	-20,491	35,291

Full valuation and sensitivity

For each complete window, add that quarter's operating profit and the following three quarters, multiply by 20 and divide by the saved starting-quarter shares. Risk and currency factors are both one. The new adverse state changes earnings once; the analytical Base and Excel Base do not multiply the ceiling.

Ceiling quarter	Exact earnings window	F4Q OP, USD M	Multiple	Shares M	Ceiling USD
Q3 2026	Q3 2026 through Q2 2027	647	20x	93	139
Q4 2026	Q4 2026 through Q3 2027	1,135	20x	96	236
Q1 2027	Q1 2027 through Q4 2027	1,693	20x	99	342
Q2 2027	Q2 2027 through Q1 2028	2,082	20x	101	412
Q3 2027	Q3 2027 through Q2 2028	2,332	20x	103	453
Q4 2027	Q4 2027 through Q3 2028	2,429	20x	105	463
Q1 2028	Q1 2028 through Q4 2028	2,373	20x	106	448

Only seven complete valuation windows are available. The operating forecast ends Q4 2028; later ceiling quarters remain unavailable. No 2029 or 2030 operating extension is invented to fill the ladder.

Calendar-year cross-check

Period	Coverage	Previous revenue, USD M	Current revenue, USD M	Previous OP, USD M	Current OP, USD M	OP change
2026	Q3 and Q4 only	777	740	98	82	-17%
2027	Full year	5,810	5,729	1,730	1,693	-2%
2028	Full year	7,365	7,365	2,373	2,373	0%

Probability and retained-cost sensitivity

The selected 25% lag probability and 15% retained displaced COGS are judgments about a bounded outcome. The alternatives below vary the same common state across affected quarters, with the other model inputs fixed. A no-increment probability reproduces the prior financial forecast exactly. The sensitivity changes no score or contract.

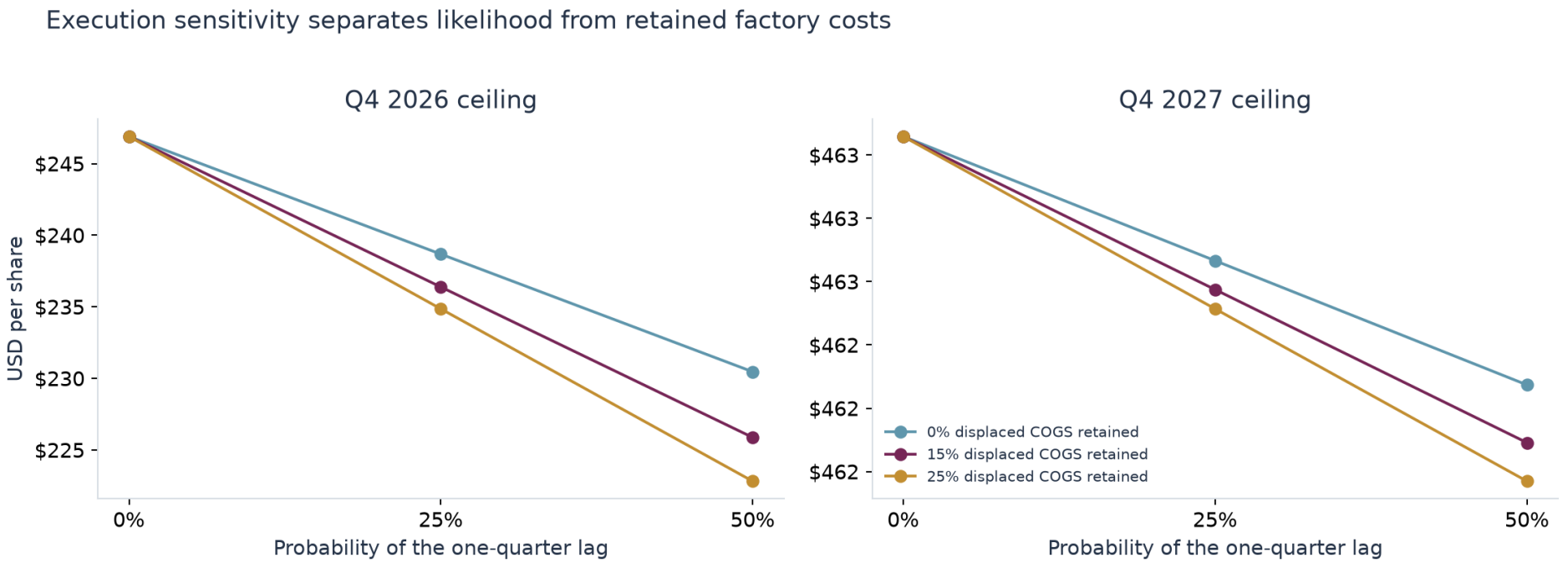


Figure 5. Execution sensitivity for earlier and active valuation windows. The common adverse state varies from 0% to 50%; retained displaced COGS varies from 0% to 25%.

Lag probability	Displaced COGS retained	Q4 2026 ceiling, USD	Q4 2027 ceiling, USD	FY2027 OP, USD M
0%	0%	247	463	1,730
0%	15%	247	463	1,730
0%	25%	247	463	1,730
25%	0%	239	463	1,701
25%	15%	236	463	1,693
25%	25%	235	463	1,688
50%	0%	230	462	1,672

Lag probability	Displaced COGS retained	Q4 2026 ceiling, USD	Q4 2027 ceiling, USD	FY2027 OP, USD M
50%	15%	226	462	1,657
50%	25%	223	462	1,646

Retained policy and dilution controls

The following changes only the policy probability, holding the current 800G case and share schedule fixed. It is a separate policy diagnostic, not another source of independent portfolio diversification.

Policy probability	Q4 2027 ceiling, USD
0%	459
25%	461
40%	463
50%	464
75%	466

Additional Q4 2027 shares	Q4 2027 ceiling, USD	Interpretation
None	463	Current 105 million economic-share allowance
5 million	442	Unselected future dilution sensitivity
10 million	422	Unselected future dilution sensitivity

Validation, counter-evidence and required next facts

The staged calculation reproduces the prior forecast when the new delay probability is zero. All eight product vectors reconcile to total revenue and GP, including the separate policy substitution residual. The selected state changes only 800G conversion, its associated COGS/GP, consolidated results and dependent rolling ceilings. Starting-quarter shares, the 20x method, currency, the reference 1.6T/ELSFP paths and existing policy increments remain unchanged. Excel Base is protected at 72; the artifact analytical Base is 70.

The most useful next evidence is a dated accepted-output series by speed, factory and customer, together with comparable installed capacity, test throughput, DSP/TIA allocation and realized unit cost. If the 10%-20% source figure uses a different denominator or reflects an already-resolved stage, reduce or remove the selected lag. If product-specific delays persist beyond one quarter or affect 1.6T, assess that new scope explicitly, including retained costs and customer substitution. More nominal floor space alone does not resolve the constraint.

For ELSFP, verify actual qualification, product power and wavelength specification, customer-order scope, modules per system, packaging yield and recognized shipment dates against the official end-2027 nominal milestone. The AWS NPO figure needs a product mapping before entering this vector. For substrate risks, compare qualified material grades and good-unit capacity while retaining competing expansion, source uncertainty and the expert/publisher disagreement.

For capital, obtain the next companywide legal count, award and Amazon vesting roll-forward, actual weighted EPS shares, and a complete factory/working-capital cash budget. A material financing or an uncovered funding need requires a new period-specific state. No automatic target uplift is assigned to cash that funds the capacity already in the model.

Source coverage: all four supplied analyst images were visually reviewed. Original report points 1-5, analyst identity, exact production observations and malformed financial-unit labels remain missing. All 16 pages of the InP call PDF were reviewed; its speaker is anonymous and it is a qualified secondary check. The AAOI official September 21 deck was read through the web PDF and page image, with direct download unavailable. Current evidence and all limitations are recorded separately from model adoption.